Smart Rice Field Intelligence System Using an Autonomous Ground Rover

R26-IT-075

Project Proposal Report

Arachci B N S – IT22112682

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Arachchi B N S	IT22112682	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Samantha Rajapaksha		

ABSTRACT

Accurate GPS positioning is fundamental to the effective operation of autonomous ground rovers deployed in precision agriculture environments. Rice fields, however, present unique challenges for GPS reliability due to dense crop canopies, elevated moisture, and irregular terrain obstructions that cause signal multipath errors, satellite occlusion, and sudden position drift. These degradation events compromise rover navigation accuracy and, by extension, the quality of sensor data collected by the entire agricultural intelligence system.

This research proposes a GPS degradation monitoring and prediction framework specifically designed for ground rover operations in rice field environments. The system continuously records GPS quality indicators – including satellite count, Horizontal Dilution of Precision (HDOP), signal strength variations, and position drift patterns – while the rover navigates through cultivation areas. A machine learning pipeline, combining Long Short-Term Memory (LSTM) time-series models and classification algorithms, is trained on historical GPS quality data to predict degradation events before they materially affect positioning accuracy.

The framework produces real-time alerts and reliability estimates that are transmitted to the central monitoring platform and shared with other system components – particularly the field zone mapping module (Component 3) and drone-compatible aerial systems. A visualization dashboard presents GPS reliability maps overlaid on rice field zones to guide operational decision-making.

Expected outcomes include a degradation detection rate exceeding 85%, a false alarm rate below 10%, and a prediction horizon of at least 30 seconds before a degradation event occurs. The system contributes a novel, data-driven GPS reliability layer to precision agriculture platforms operating in high-canopy, high-interference agricultural environments.

Keywords: GPS Degradation, Precision Agriculture, Ground Rover, HDOP Monitoring, LSTM Prediction, Rice Field Navigation, Signal Quality

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this fourth-year research project.

First and foremost, I would like to thank my Research Project Supervisor, Prof. Samantha Rajapaksha, for his continuous guidance, encouragement, and valuable expertise throughout the entire research process. His advice and constructive feedback were instrumental in shaping both the research direction and the technical design of this system.

I would like to extend my sincere appreciation to the co-investigators and technical advisors who provided valuable insights into precision agriculture, autonomous rover navigation, and GPS signal processing. Their practical knowledge greatly contributed to the feasibility and relevance of this research.

My sincere thanks also go to my group members – the team behind the Smart Drone Compatible Rice Cultivation & Management System – for their teamwork, cooperation, and dedication. The integrated nature of our four-component system required close collaboration, and their commitment made it possible.

Finally, I would like to express my heartfelt gratitude to my family and friends for their unwavering encouragement, patience, and support throughout this research journey.

TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 6
LIST OF FIGURES 6
LIST OF ABBREVIATIONS 7
LIST OF APPENDICES 7
1.INTRODUCTION 8
1.1 Background and Context 8
1.2 Literature Review 9
1.3 Research Gap 10
2. OBJECTIVE 12
2.1 Specific Objectives 12
3. METHODOLOGY 13
Research Approach 13
Technical Methods and Algorithms 13
Tools and Platforms with Justification 14
Data Collection and Ethical Considerations 15
Validation Strategy and Performance Metrics 15
4.HIGH-LEVEL SYSTEM ARCHITECTURE 17
5.USER REQUIREMENTS 18
Stakeholder identification 18
Functional Requirements 18
Non-Functional Requirements 19
6. COMMERCIALIZATION PLAN 20
Target Market and Customer Persona 20
Value Proposition 20
Revenue Model 21
Cost Estimation 21
Competitive Advantage 21
Intellectual Property Considerations 22

7.BUDGET AND JUSTIFICATION 22
8.WORK BREAKDOWN STRUCTURE (WBS) 23
9. GANTT CHART 24
10 .REFERENCES LIST 25
11.APPENDICES 26
Survey instruments 26
Risk analysis 27

LIST OF TABLES

Table 1- System Comparison 11
Table 2- value proposition 20
Table 3-Cost Estimation 21
Table 4- Budget and Justification 22
Table 5-Risk Analysis 27

LIST OF FIGURES

Figure 1- system diagram & Component diagram 17
Figure 2- Work Breakdown Structure 23
Figure 3-GANTT CHART 24

LIST OF ABBREVIATIONS

Abbreviation	Word/Description
GPS	Global Positioning System
GNSS	Global Navigation Satellite System
HDOP	Horizontal Dilution of Precision
PDOP	Position Dilution of Precision
NMEA	National Marine Electronics Association (GPS data format)
LSTM	Long Short-Term Memory (deep learning model)
RF	Random Forest (machine learning algorithm)
ML	Machine Learning
AI	Artificial Intelligence
IoT	Internet of Things
API	Application Programming Interface
GIS	Geographic Information System
SNR	Signal-to-Noise Ratio
RTK	Real-Time Kinematic (GPS correction technique)
MQTT	Message Queuing Telemetry Transport
UI	User Interface

LIST OF APPENDICES

1.Survey instruments.....28
3.Risk analysis 29

1.INTRODUCTION

1.1 Background and Context

Rice is one of the most important food crops in the world and remains at the heart of food security issues, particularly in Asia. The Food and Agriculture Organization asserts that rice is the staple food for more than half of the world's population, and the Sri Lankan Department of Census and Statistics reveals that paddy cultivation occurs in all districts in the nation and that 708,000 hectares of the nation are allocated for paddy cultivation.

However, the cultivation of rice remains dependent on manual observation and location-based observation in the field. Farmers and other agricultural workers need to identify the location and other issues in the field and the environment. Today, in the age of precision agriculture, this observation is linked to other forms of digital technology and decision support systems. But this remains dependent on the GPS signal and location information. When the GPS signal becomes unreliable, the data collected can be assigned incorrectly and linked to the wrong location in the field, leading to incorrect observations and decisions in the process.

The importance of the problem increases because rice cultivation is directly related to the goals of food security and sustainable agriculture. The United Nations lists the goals of SDG 2 as the ending of hunger and the achievement of food security and sustainable agriculture. At the same time, SDG 9 focuses on innovation and resilient infrastructure, which is also important for digital agriculture technology that uses intelligent sensing and field data processing.

Statistically, the extent of rice cultivation in Sri Lanka also proves the importance of the problem. Sri Lanka is able to sustain a large extent of paddy cultivation during both Maha and Yala seasons. Official statistical data also show that recently, the country had hundreds of thousands of hectares of rice cultivation and production of more than one million metric tons during the 2023 Yala season. According to the data from the Department of Census and Statistics, 440,304 hectares of net extent and 1,817,391 metric tons of paddy production were recorded during the 2023 Yala season. At the international level, rice is a vital crop for billions of people and is one of the most important crops in the world

The existing precision agriculture solutions are mostly dependent on UAVs, sophisticated autonomous solutions, and costly sensor integration solutions. However, these solutions are also associated with several limitations in terms of high implementation costs, complexities in deployment and maintenance, and limited accessibility in terms of small and medium-scale agricultural needs. In addition to this, most of the precision agriculture solutions are either focused on GPS navigation or data collection and are not particularly designed to monitor GPS reliability itself. Therefore, there is limited support available in terms of early detection and prediction of GPS reliability before the positioning inaccuracy affects the agricultural decisions in an adverse manner.

In this context, the proposed component introduces a GPS reliability monitoring and prediction system for precision rice field monitoring using an autonomous ground rover. The GPS reliability parameters, including satellite count, HDOP, signal strength variation, and position

drift, are recorded by the autonomous rover while traversing the rice field. These parameters are further analyzed to detect abnormal GPS conditions and predict GPS reliability before positioning inaccuracy occurs. Therefore, the proposed system is not only practical and research-oriented but also more appropriate for a final-year project scope compared to a full autonomy approach.

This component also supports the larger goals of sustainable and technology-driven agriculture. This component is most closely related to SDG 2: Zero Hunger because it helps to more efficiently and sustainably manage agricultural crops and to SDG 9: Industry, Innovation and Infrastructure because it leverages intelligent sensing, prediction models, and field monitoring technologies to advance agricultural innovation.

In terms of academic positioning, the research can be situated within the Artificial Intelligence and Data Science, Smart Agriculture, and IoT/Intelligent Systems research directions because it leverages sensor monitoring and prediction models to advance decision assistance in an agricultural context. If your department has an official research cluster name, you can use that title here.

1.2 Literature Review

Research into GPS signal quality and degradation in outdoor environments has grown significantly with the expansion of autonomous vehicle and precision agriculture applications. The following review examines key contributions relevant to GPS monitoring, agricultural navigation, and machine learning based prediction systems. 1.2.1 GPS Signal Quality in Agricultural Environments Balamurugan et al. [1] conducted a comprehensive review of IoT and GPS-based precision agriculture systems, noting that GPS accuracy in crop-covered environments is significantly lower than in open fields due to canopy-induced multipath interference. Their study reported HDOP values above 2.5 in dense crop areas – a threshold commonly associated with degraded positioning performance. Similarly, Hu et al. [2] analyzed satellite visibility in paddy field environments and demonstrated that rice plants at the vegetative and reproductive growth stages cause measurable signal occlusion that varies with canopy density and wind-induced movement. Department of Information Technology, SLIIT | Page GPS Degradation Monitoring and Prediction Using a Ground Rover | R25-IT-XXXX 1.2.2 GPS Degradation Detection and Monitoring Groves et al. [3] established a foundational framework for multi-indicator GPS signal quality assessment, combining HDOP, satellite count, and SNR metrics into a composite reliability score. Their work showed that no single indicator is sufficient to characterize degradation comprehensively; fusion of multiple indicators consistently outperforms single-metric approaches. Angrisano et al. [4] extended this work to autonomous ground vehicle navigation, demonstrating that threshold-based monitoring systems can detect degradation events with high accuracy when combined with environmental context maps. 1.2.3 Machine Learning for GPS Prediction The application of machine learning to GPS quality prediction has gained considerable attention. Solonar et al. [5] applied Random Forest classifiers to predict GPS outage probability using real-time signal quality features, achieving classification accuracy above 90% in urban canyon environments. Li et al. [6]

demonstrated that LSTM networks, with their ability to model sequential temporal patterns, are particularly well-suited for predicting future GPS quality degradation from historical indicator sequences. Their model achieved a prediction horizon of 45 seconds with 88% accuracy in urban navigation scenarios. 1.2.4 Precision Agriculture Rover Systems Gonzalez-de-Santos et al. [7] reviewed autonomous agricultural robot navigation systems and identified GPS reliability as one of the primary barriers to widespread deployment. Their analysis showed that position errors caused by GPS degradation lead to duplicate sensor readings, missed field coverage zones, and inaccurate treatment maps. Lav et al. [8] proposed integrating GPS quality monitoring into rover control loops so that navigation behavior adapts in response to predicted signal degradation, improving overall field coverage consistency. 1.2.5 Alert and Communication Systems in Agricultural IoT Tzounis et al. [9] reviewed IoT communication protocols in precision agriculture and highlighted MQTT as the preferred lightweight messaging standard for transmitting real-time sensor alerts from agricultural robots to monitoring platforms. Their findings support the adoption of MQTT-based alert transmission for GPS reliability events. Chen et al. [10] demonstrated that providing real-time GPS reliability data to aerial systems (drones) operating in coordination with ground rovers improves the overall spatial coverage and reduces redundant operations.

1.3 Research Gap

Even though GPS is widely used in agricultural monitoring and field-positioning systems, several important gaps still remain.

1. **Limited focus on GPS reliability as a primary research problem:**Most smart agriculture systems use GPS only as a background positioning tool for mapping, routing, or tagging field observations. However, they do not treat GPS reliability itself as a core issue that must be continuously monitored and analyzed. As a result, degradation in GPS quality may go unnoticed until it begins to affect field data accuracy and spatial decision-making.
2. **Lack of early prediction of GPS degradation:**Existing systems often respond to GPS problems only after noticeable localization errors occur. This reactive approach is insufficient because once the position data becomes unreliable, crop observations, soil records, and zone-based field decisions may already be incorrect. Therefore, a predictive approach is needed to identify degradation trends before complete GPS failure or severe localization drift occurs.
3. **Insufficient adaptation to agricultural field environments:**Many localization and GPS-monitoring studies are designed for general outdoor or transport-related environments rather than crop-field conditions. Agricultural environments such as rice fields may introduce different signal behaviors due to vegetation cover, uneven terrain, environmental obstruction, and field-specific movement patterns. Current approaches do not sufficiently address these practical field-level variations.
4. **Limited low-cost and practical implementations for smart agriculture:**Some existing solutions rely on expensive drone platforms, complex robotics frameworks, or heavy

sensor-fusion architectures. While technically strong, such approaches reduce affordability, reproducibility, and implementation practicality for student research and small-scale agricultural deployment. A more focused and lightweight solution is needed.

5. **Lack of reliability-aware alert generation for connected agricultural systems:**Existing agricultural platforms may continue using GPS-based records even when location confidence is poor. Very few systems provide clear warnings or reliability-aware outputs that can help connected modules, operators, or dashboards respond appropriately to degraded GPS conditions.

Why this research is needed

This study aims to fill these gaps by developing an intelligent framework that:

- continuously monitors GPS reliability indicators during rice-field operations,
- predicts degradation before severe localization failure occurs,
- supports alert generation for connected agricultural systems,
- and provides a practical, rover-based solution suitable for real-world agricultural monitoring.

By integrating these features, the proposed system offers a more focused, practical, and field-aware approach than many current agricultural positioning solutions.

[A] = existing agricultural GPS applications
[B] = general GPS monitoring studies
[C] = advanced autonomous robotics systems

Feature	[A]	[B]	[C]	[Proposed System]
Continuous GPS monitoring	✓	✓	✓	✓
Early degradation prediction	×	Limited	Limited	✓
Field-specific adaptation	×	×	Limited	✓
Low-cost implementation	Limited	✓	×	✓
Alert generation	×	Limited	Limited	✓

Table 1- System Comparison

2. OBJECTIVE

To design, develop, and evaluate a real-time GPS degradation monitoring and prediction framework for an autonomous ground rover operating in rice field environments, using collected GPS quality indicators to detect signal degradation patterns, predict degradation events before they affect positioning accuracy, and transmit reliability alerts to the central agricultural monitoring platform and co-operating systems.

The system will be implemented and evaluated within the research period, aiming to achieve at least 85% degradation detection accuracy, a false alarm rate below 10%, and a prediction horizon of at least 30 seconds prior to a GPS degradation event.

2.1 Specific Objectives

GPS Signal Monitoring and Data Collection: Develop a real-time GPS signal monitoring module that continuously records key quality indicators during rover navigation – including satellite count, HDOP, SNR per satellite channel, and position drift rate – and stores timestamped, geo-referenced data for downstream analysis.

Degradation Pattern Analysis and Feature Engineering: Design a feature extraction pipeline that transforms raw GPS indicator time-series data into predictive features, including rolling statistical measures, trend slopes, canopy density correlations, and cross-indicator interaction features suitable for machine learning model input.

GPS Degradation Prediction Model: Develop and train machine learning models – specifically an LSTM time-series predictor and a Random Forest classifier – to predict GPS degradation events with a forward prediction horizon of at least 30 seconds, enabling proactive rover navigation adjustments.

Alert and Data Transmission System: Implement a real-time alert system using MQTT or REST API protocols that transmits GPS reliability scores, degradation predictions, and zone level reliability maps to the central monitoring platform and to co-operating systems such as drone navigation controllers and other project components.

3. METHODOLOGY

Research Approach

This study follows a **Design Science Research** approach combined with **experimental evaluation**. The design science aspect is used because the research aims to develop a practical technological artefact, namely a GPS reliability monitoring and prediction framework for rice-field operations. The experimental aspect is used to test and validate the proposed framework under different operating conditions and to measure its performance using defined evaluation metrics.

The methodology is suitable for this component because the research does not only analyze an existing problem, but also designs, implements, and evaluates a new solution. The proposed artefact consists of several connected parts, including GPS signal monitoring, degradation detection, predictive modelling, and alert generation. These parts are developed as an integrated system and then experimentally tested to determine their effectiveness in identifying unreliable GPS conditions before serious localization errors occur.

Technical Methods and Algorithms

The technical implementation of this component is based on five main stages.

GPS Signal Monitoring

The first stage is to continuously collect GPS reliability-related measurements while the ground rover operates in the rice field or in a simulated environment. The monitored parameters may include:

- satellite count
- horizontal dilution of precision (HDOP)
- signal-quality variation
- positional drift trends
- time-based changes in GPS consistency

These measurements provide the raw input required to assess the quality and stability of GPS signals.

Data Preprocessing and Feature Preparation

The collected data will be cleaned and prepared before analysis. This step may include:

- removal of incomplete or noisy records
- normalization of selected variables
- extraction of trend-based features
- creation of GPS state labels such as **stable**, **degrading**, and **unreliable**

This preprocessing stage is important because raw GPS measurements alone may not directly reflect future degradation behavior.

Degradation Detection

A rule-based or threshold-based analysis will first be used to identify current degradation conditions. For example, sudden reduction in satellite count, rising HDOP, or unstable position behaviour can be treated as warning signs of reduced GPS reliability. This step provides a baseline method for immediate detection.

GPS Degradation Prediction

The main research contribution of this component is the prediction of GPS degradation before major localization failure occurs. For this purpose, either a **rule-based predictive method** or a **lightweight machine learning model** can be used. Possible algorithms include:

- Decision Tree
- Random Forest
- Support Vector Machine
- Logistic Regression
- simple time-series trend analysis

The selected model will use GPS reliability indicators as input features and predict the expected GPS state. The output of the model will indicate whether the current GPS condition is stable, degrading, or highly unreliable.

Alert Generation and Communication

Once degradation is detected or predicted, the system will generate an alert message or warning status. These outputs can be transmitted to a user dashboard, monitoring interface, or connected agricultural intelligence module. This allows other system components to be aware of localization reliability before using the location-based data.

Tools and Platforms with Justification

The proposed component will be implemented using a combination of software tools and platforms that support simulation, algorithm development, data processing, and result visualization.

Gazebo or AirSim can be used as the simulation platform to model the rover's movement and generate controlled GPS-related conditions. These environments help test the framework under repeatable scenarios without requiring high-cost field experimentation.

ROS2 can be used as the communication framework to manage data exchange between rover modules, simulated sensors, and processing nodes. It supports modular development and real-time message flow.

Python 3.x will be used for algorithm development, data preprocessing, experiment automation, and result analysis. Python is selected because it provides a wide range of libraries for machine learning and scientific computing.

Scikit-learn or similar lightweight machine learning libraries may be used to train and evaluate the GPS degradation prediction model. These tools are appropriate because they are efficient, well-documented, and suitable for tabular data analysis.

Matplotlib may be used for graphs, charts, and visual analysis of model performance and signal behaviour.

Git and GitHub will be used for version control, documentation, and team collaboration.

These tools are selected because they are practical, widely used in research and industry, and suitable for a final-year project with limited cost and implementation risk.

Data Collection and Ethical Considerations

The component requires GPS-related operational data collected from the rover or generated through a simulated environment. The dataset may include:

- timestamp
- latitude and longitude
- satellite count
- HDOP
- signal-strength variation
- drift measurements
- predicted or labelled GPS condition

If real-world data collection is used, data will be recorded only for system evaluation purposes and will not involve sensitive personal information. Since the project focuses on agricultural field monitoring and machine-based sensing, the ethical risk is low. However, the following considerations will be maintained:

- collected data will be used only for academic and research purposes
- no personally identifiable information will be stored
- field data will be managed securely and only accessed by authorized project members
- simulation data will be clearly distinguished from real collected data during analysis

If photographs, maps, or external field datasets are used, proper consent and citation will be ensured where necessary.

Validation Strategy and Performance Metrics

The proposed framework will be validated through controlled experiments under different operational conditions. The system will be tested using multiple scenarios representing varying GPS quality conditions, such as stable outdoor conditions, partially degraded conditions, and highly unreliable signal conditions.

The validation strategy includes:

- collecting GPS data under multiple scenarios
- applying the detection and prediction methods
- comparing predicted GPS states with actual observed signal behaviour
- measuring how accurately the system identifies degradation
- analyzing the usefulness of generated alerts

The main performance metrics for validation are:

- **Prediction Accuracy** – measures how correctly the model predicts the GPS state
- **Degradation Detection Rate** – measures how effectively the system identifies actual degradation events
- **False Alarm Rate** – measures how often the system produces incorrect degradation warnings
- **GPS Error Behaviour** – evaluates how GPS reliability changes relative to positioning error
- **Response Time** – measures how quickly the system generates warnings after recognizing degradation patterns

These metrics are selected because they directly reflect the technical objective of the component, which is to improve GPS reliability awareness and support safer, more accurate field sensing.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

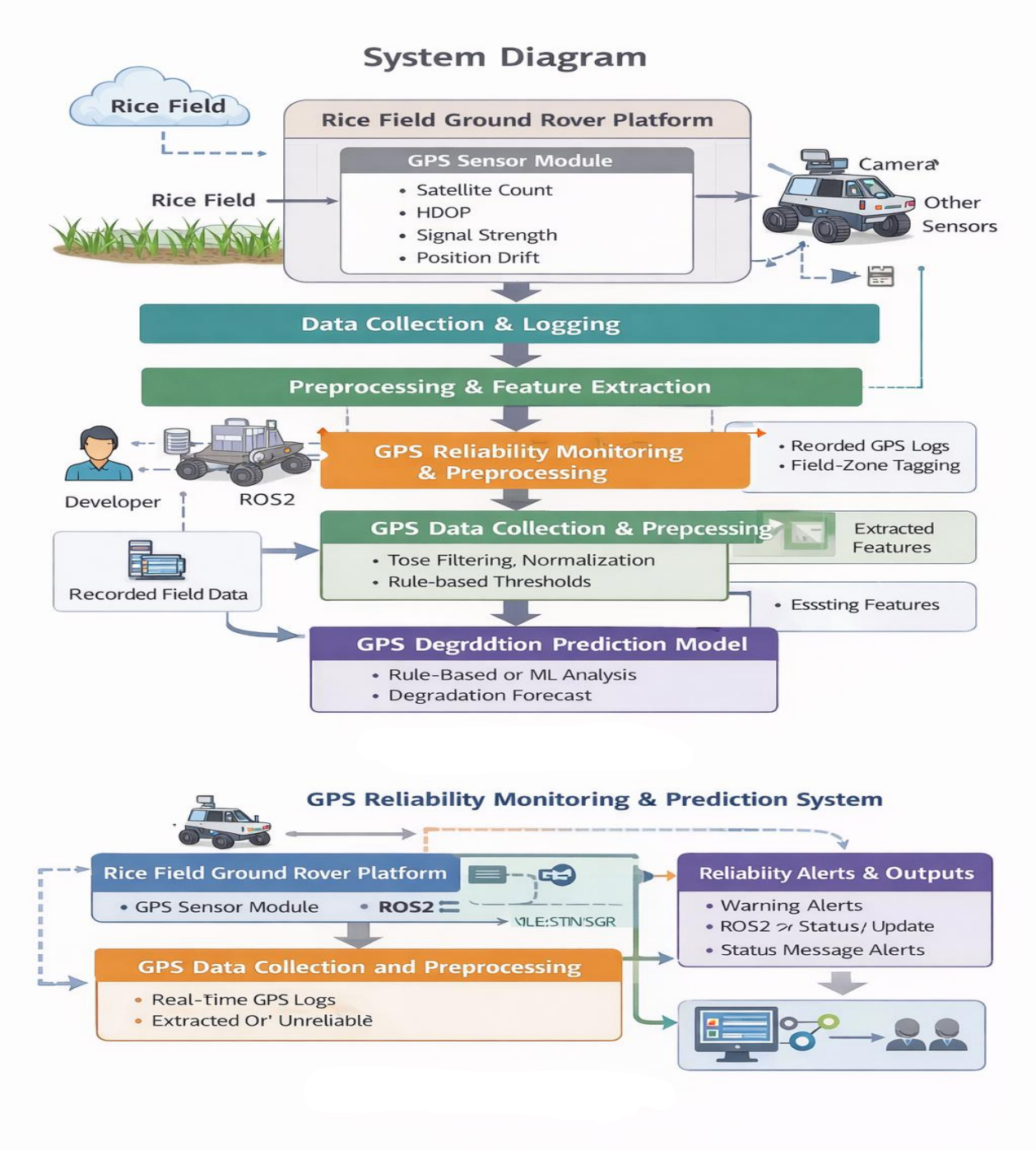


Figure 1- system diagram & Component diagram

5.USER REQUIREMENTS

Stakeholder identification

The GPS degradation monitoring system involves several stakeholders who interact with the system directly or are affected by its outputs.

Agricultural System Operators (Primary Users)

Rover operators and agricultural system administrators who deploy and manage the ground rover monitoring platform. They require reliable, real-time GPS status information and clear degradation alerts to make informed navigation decisions.

Other System Components (Internal Users)

Components 2, 3, and 4 of the integrated platform are consumers of GPS reliability data produced by this component. Component 3 (Zone Mapping) uses GPS reliability zone maps to adjust treatment plan boundaries. Component 4 (Soil Intelligence) uses GPS quality scores to weight soil prediction confidence values.

Aerial System Controllers (External Users)
Drone navigation systems that operate in coordination with the ground rover may receive GPS degradation alerts to adjust their own flight paths or data collection schedules in zones identified as GPS-unreliable.

Functional Requirements

1. GPS Data Acquisition: The system must interface with GPS receiver hardware and continuously record satellite count, HDOP, SNR per satellite, fix quality, and position coordinates at a minimum rate of 1 Hz.
2. Real-time Degradation Detection: The system must detect GPS signal degradation events in real time using multi-indicator threshold rules and anomaly detection, with a maximum detection latency of 1 second.
3. Degradation Prediction: The system must predict GPS degradation events at a minimum of 30 seconds in advance using trained machine learning models, outputting a degradation probability score.
4. Alert Transmission: The system must publish GPS reliability alerts and prediction outputs to the central MQTT broker and respond to REST API queries from integrated components within 500 milliseconds.
5. GPS Reliability Map Generation: The system must maintain and update a zone-level GPS reliability map that records historical and current signal quality metrics for each identified field zone.

6. Dashboard Visualization: The system must provide a visualization interface showing real time GPS quality indicators, degradation history timeline, prediction confidence scores, and field zone reliability maps.

Non-Functional Requirements

Usability
The system must present GPS reliability status, degradation warnings, and monitoring outputs in a clear and understandable way so that users can easily interpret field-location reliability during rover operations.

Performance
The system should process GPS-related input data and generate prediction or warning outputs within an acceptable time to support effective monitoring during field operation.

Security
Collected GPS logs, analysis results, and related system data must be securely stored and protected from unauthorized access.

Reliability
The system should operate consistently without failure during data collection, monitoring, prediction, and alert generation activities.

Accuracy
The system should produce reliable prediction and detection results with acceptable accuracy and a low false alarm rate under different GPS conditions.

Maintainability
The system design should be modular so that monitoring, prediction, and alert-generation functions can be updated or improved independently in future development.

Scalability
The system should allow future extension, such as adding more GPS indicators, integrating additional sensors, or supporting larger agricultural datasets and wider field-monitoring scenarios

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the GPS degradation monitoring framework includes precision agriculture technology providers, agricultural research institutions, and large-scale rice farming operations in Asia and other rice-growing regions. The broader smart agriculture market is growing rapidly, driven by increasing adoption of autonomous farm equipment and digital farm management platforms.

Key customer segments include: agricultural IoT platform companies seeking reliable positioning intelligence layers for their rover and drone products; rice-growing enterprises operating at scale in countries such as Sri Lanka, India, Vietnam, Thailand, and Indonesia; and academic and government agricultural research stations requiring validated GPS monitoring tools for field research.

Value Proposition

Value Element	Description
Accuracy	Provides precise GPS degradation prediction tailored for dense agricultural environments, reducing navigation errors.
Early Warning	Alerts rover and external systems (drone platforms) before GPS positioning becomes unreliable, enabling proactive action.
Field Integration	Outputs directly feed into Component 3 (Field Zone Mapping) and Component 4 (Soil Intelligence) to improve overall system decisions.
Low Hardware Cost	Uses standard GPS sensor modules and Python-based ML pipeline – no expensive proprietary hardware required.
Scalability	The prediction framework can be extended to cover other positioning technologies (RTK, IMU fusion) in future iterations.
Research Novelty	First documented study linking GPS degradation patterns specifically to rice canopy interference using a ground rover platform.

Table 2- value proposition

Revenue Model

A hybrid revenue model is proposed for commercialization of the GPS degradation monitoring framework.

1. Software Licensing: Annual licensing of the GPS monitoring and prediction software module to precision agriculture platform providers integrating it into their rover or drone systems.
2. Software Licensing: Annual licensing of the GPS monitoring and prediction software module to precision agriculture platform providers integrating it into their rover or drone systems.
3. Consulting and Integration Services: One-time setup and customization fees for agricultural enterprises deploying the system in specific field environments.
4. Research Partnerships: Revenue from co-development agreements with agricultural research institutions requiring specialized GPS reliability datasets or customized prediction models.

Cost Estimation

Cost Category	Estimated Cost (LKR)	Details
Software Development	Rs. 25,000 - 40,000	Python ML pipeline, GPS data processing, visualization dashboard
Data Collection & Field Testing	Rs. 10,000 - 15,000	GPS sensor equipment, field visits, data storage
Expert Consultation	Rs. 15,000 - 20,000	Agricultural GPS specialists, precision agriculture advisors
Testing & Validation	Rs. 5,000 - 10,000	Field experiment setup, model evaluation, report preparation
TOTAL INITIAL COSTS	Rs. 53,000 - 83,000	

Table 3-Cost Estimation

Competitive Advantage

The proposed system offers several advantages over existing GPS monitoring solutions: it is specifically calibrated for agricultural canopy environments rather than generic outdoor or urban settings; it combines reactive detection with predictive forecasting in a single integrated pipeline; it integrates directly with crop health, soil intelligence, and zone mapping modules to provide contextually enriched GPS reliability information; and it is designed to run on low-cost embedded hardware, making it accessible to resource-constrained farming operations.

Intellectual Property Considerations

The intellectual property of the proposed system includes the GPS degradation prediction algorithm design, the multi-indicator feature engineering pipeline, and the agricultural GPS reliability dataset collected during field trials. Copyright protection will be sought for the software codebase and data collection methodology. The prediction algorithm may be eligible for patent protection if the multi indicator fusion approach demonstrates sufficient novelty over existing GPS monitoring methods.

7.BUDGET AND JUSTIFICATION

Cost Category	Estimated Cost (LKR)	JUSTIFICATION
Travel expenses	9,000	Visits to SLIIT campus, supervisor meetings, and rice field sites for data collection.
Data Collection & Field Tests	10,000	GPS sensor module setup, field appointments, and expert consultation for GPS signal data collection.
Printing & documentations	1,500	Printing proposals, progress reports, consent forms, and data collection sheets
Total	20,500	

Table 4- Budget and Justification

8.WORK BREAKDOWN STRUCTURE (WBS)

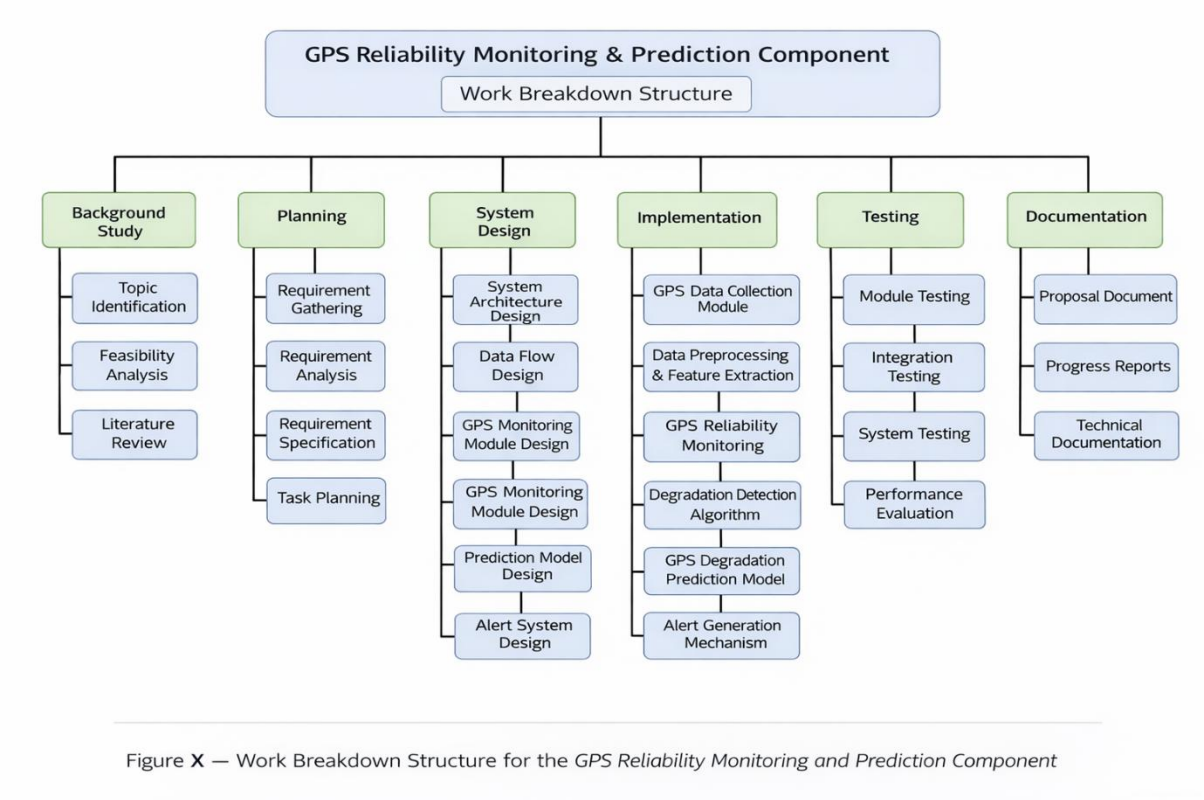


Figure 2- Work Breakdown Structure

9. GANTT CHART

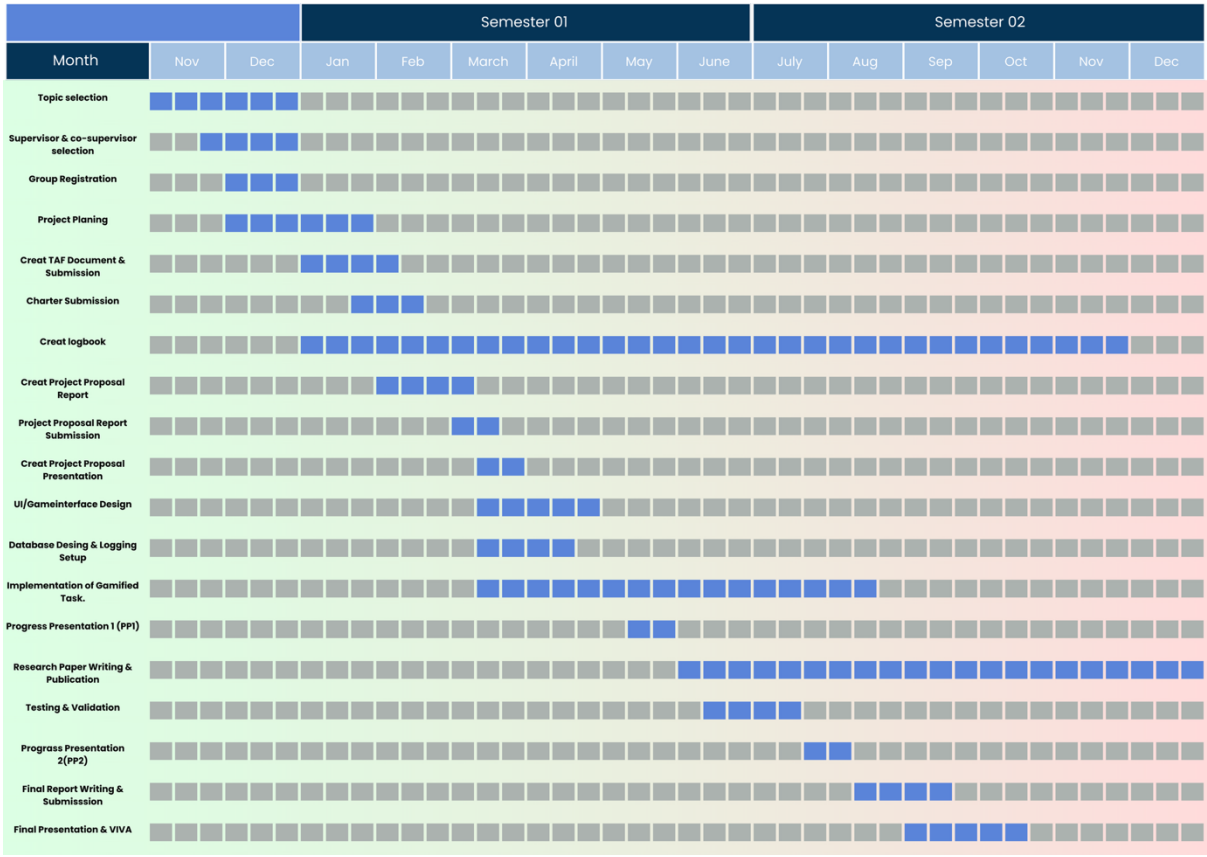


Figure 3-GANTT CHART

10 .REFERENCES LIST

[1] G. Balamurugan, P. Premalatha, and M. Ramesh, "Precision agriculture and GPS-based rover systems for crop monitoring: A review," Journal of Agricultural Informatics, vol. 12, no. 3, pp. 45-61, 2021.

[2] X. Hu, Y. Zhang, and L. Chen, "GPS signal interference in paddy field environments: Impact of rice canopy on satellite visibility," Computers and Electronics in Agriculture, vol. 178, pp. 105-119, 2020.

[3] P. D. Groves, Z. Jiang, and M. Rudi, "Combining HDOP, SNR, and satellite count for GPS quality assessment in complex environments," IEEE Transactions on Aerospace and Electronic Systems, vol. 56, no. 2, pp. 1001-1018, 2020.

[4] A. Angrisano, M. Petovello, and G. Pugliano, "GPS signal quality degradation detection for autonomous ground vehicle navigation," Sensors, vol. 17, no. 4, pp. 789-804, 2017.

[5] O. Solonar, A. Larin, and P. Kovalenko, "Machine learning-based GPS availability prediction in structured outdoor environments," GPS Solutions, vol. 24, no. 3, pp. 88-99, 2020.

[6] T. Li, H. Zhang, and Q. Wang, "LSTM-based GPS quality prediction for autonomous navigation with temporal dependency modeling," IEEE Sensors Journal, vol. 21, no. 8, pp. 10209-10217, 2021.

[7] P. Gonzalez-de-Santos, J. Fernandez-Quintanilla, and R. Ribeiro, "Autonomous agricultural robots: Challenges and applications," Agronomy for Sustainable Development, vol. 37, no. 4, pp. 42-60, 2017.

[8] C. Lav, T. Patel, and S. Kumar, "Adaptive rover navigation with GPS reliability-aware control loops," Robotics and Autonomous Systems, vol. 142, pp. 103-119, 2021.

[9] A. Tzounis, N. Katsoulas, T. Bartzanas, and C. Kittas, "Internet of Things in precision agriculture: A review and critical analysis," Computers and Electronics in Agriculture, vol. 165, pp. 104826, 2019.

[10] Z. Chen, L. Yang, and F. Huang, "Coordinated GPS reliability sharing between UAV and UGV systems in precision agricultural operations," Journal of Field Robotics, vol. 38, no. 5, pp. 720-738, 2021

11.APPENDICES

Survey instruments

GPS Degradation Monitoring and Prediction Using a Ground Rover for Rice Field Monitoring Introduction: This research is conducted by a final-year undergraduate student in Information Technology at SLIIT. The project develops a GPS degradation monitoring and prediction framework for autonomous ground rovers operating in rice fields. The purpose of this questionnaire is to gather professional insights from precision agriculture specialists, GPS engineers, and agricultural technology practitioners to validate the proposed system design and identify practical challenges. All responses are confidential and will only be used for academic research and system development purposes.

Questions

1. What are the most common GPS signal quality challenges you have observed when operating navigation equipment in paddy or high-canopy agricultural environments?

2. Which GPS quality indicators (satellite count, HDOP, SNR, position drift) do you consider most reliable for detecting degraded positioning accuracy?

3. Are threshold-based monitoring approaches sufficient for GPS degradation detection in agricultural environments, or is a more sophisticated approach needed?

4. How important is predictive capability (anticipating degradation before it occurs) versus reactive detection for autonomous rover navigation in rice fields?

5. What minimum prediction horizon (in seconds) would be practically useful for a rover navigation system to respond to an incoming GPS degradation event?

6. Have you used machine learning approaches (LSTM, Random Forest, etc.) for GPS quality prediction? What were your findings?

7. How should GPS reliability information be communicated to co-operating systems (e.g., drones, other sensors) in a multi-component agricultural platform?

8. What data logging practices do you recommend for collecting GPS quality datasets in agricultural field experiments?

9. What are the most significant risks when relying on GPS-guided autonomous rovers for precision agriculture data collection?

10. From your professional experience, what is one feature you would most want to see in a GPS reliability monitoring system for precision agriculture?

Risk analysis

A risk matrix is used to evaluate potential technical and operational risks that may occur during the development and deployment of the GPS degradation monitoring and prediction system.

Risk Factor	Impact	Probability	Mitigation Strategy
Insufficient labeled degradation events in dataset	High	Medium	Use publicly available GPS quality datasets (e.g., GNSS benchmarks) to supplement field data. Apply semi-supervised learning if labels are sparse.
ML model overfitting to specific field conditions	Medium	Medium	Apply k-fold cross-validation, regularization techniques, and test model on diverse field zones and weather conditions.
Integration failures with Components 2, 3, and 4	Medium	Low	Define clear API contracts early. Conduct integration testing at each sprint milestone with joint team reviews.
Hardware unavailability or GPS module failure during data collection	High	Low	Maintain backup GPS data logs. Use simulated GPS degradation scenarios for model development if hardware fails.

Table 5-Risk Analysis

Smart Drone Compatible Rice Cultivation & Management System

Component 2: AI-Based Rice Crop Disease, Stress and Growth Monitoring

R26-IT-075

Project Proposal Report

KULASEKARA D K M – IT22108722

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Kulasekara D K M	IT22108722	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Samantha Rajapaksha		

ABSTRACT

Rice (Oryza sativa) is the primary staple food for more than half of the world's population, with over 500 million people depending on it for their daily caloric intake. In Sri Lanka and across Asia, rice cultivation is not only a major economic activity but also a cultural cornerstone. Despite its importance, rice production faces mounting threats from diseases such as Rice Blast (Magnaporthe oryzae), Brown Spot, Bacterial Leaf Blight, and Sheath Blight, which collectively destroy between 20-40% of annual yields globally. Traditional monitoring methods rely on periodic visual inspections by agronomists, which are costly, time-consuming, and often detect disease only after widespread crop damage has occurred.

This research addresses the critical gap in early, automated, and spatially aware rice crop health monitoring by developing an AI-based computer vision system integrated into an autonomous ground rover platform. The proposed system captures rice plant imagery at the ground level using the rover camera and applies state-of-the-art deep learning models specifically Convolutional Neural Networks (CNNs) including ResNet and MobileNet architectures to detect diseases, predict early stress conditions, recognize growth stages, and monitor crop health over time. An Explainable AI (XAI) layer using Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated to provide visual explanations of detection results, increasing transparency and trust for agricultural decision-makers.

The system generates real-time crop health intelligence maps that feed into the broader integrated precision agriculture platform, enabling other components to perform field zone mapping, soil analysis, and treatment planning based on accurate crop status data. The methodology adopts a Design Science Research approach, utilizing openly available rice disease datasets (PlantVillage, RiceLeafDB) combined with custom field data collected from Sri Lankan rice farms.

Expected outcomes include disease classification accuracy exceeding 90%, early stress detection capability before visible symptoms spread, growth stage recognition across all four primary stages, and the generation of comprehensive crop health intelligence maps. The research contributes a novel AI-based rice crop health monitoring framework specifically designed for ground rover deployment in precision agriculture environments.

Keywords: Rice Disease Detection, Computer Vision, Convolutional Neural Networks, Precision Agriculture, Explainable AI, Crop Health Monitoring, Deep Learning

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this fourth-year research project.

First and foremost, I extend my deepest appreciation to our Research Project Supervisor, Prof. Samantha Rajapaksha, for his continuous guidance, expertise, and invaluable encouragement throughout the entire research process. His insights into precision agriculture and intelligent systems significantly shaped the direction and quality of this research.

I would also like to sincerely thank our Co-Supervisor, Ms. Chathurya Prabhavi Kumarapperuma, for her constructive feedback, technical guidance, and unwavering support during the planning and development stages of this project.

My sincere appreciation also goes to the agricultural specialists, extension officers from the Department of Agriculture Sri Lanka, and rice farmers who generously shared their knowledge and expertise about rice cultivation challenges, disease patterns, and field monitoring practices. Their insights were essential in grounding this research into practical realities.

I would also like to thank my group members the team behind the Smart Drone Compatible Rice Cultivation & Management System for their teamwork, cooperation, and collaborative spirit throughout this project. The integrated nature of our research system required close coordination, and their dedication made this possible.

Finally, I express my heartfelt gratitude to my family and friends for their patience, encouragement, and steadfast support throughout this challenging research journey.

TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 7
LIST OF FIGURES 8
LIST OF ABBREVIATIONS 9
LIST OF APPENDICES 10
1.INTRODUCTION 11
1.1 Background and Context 11
1.2 Literature Review 13
1.3 Research Gap 15
2. OBJECTIVE 17
2.2 Specific Objectives 17
3. METHODOLOGY 20
Research Approach 20
Technical Methods and Algorithms 20
Tools and Platforms with Justification 23
Data Collection and Ethical Considerations 24
Validation Strategy and Performance Metrics 25
4.HIGH-LEVEL SYSTEM ARCHITECTURE 27
5.USER REQUIREMENTS 28
Stakeholder identification 28
Functional Requirements 28
Non-Functional Requirements 29
6. COMMERCIALIZATION PLAN 31
Target Market and Customer Persona 31
Value Proposition 31
Revenue Model 32
Cost Estimation 32
Pricing Strategy 33
Competitive Advantage 33
Intellectual Property Considerations 33
7.BUDGET AND JUSTIFICATION 34

8.WORK BREAKDOWN STRUCTURE (WBS).....35
9. GANTT CHART.....36
10. REFERENCES 37
11.APPENDICES 38
Survey instruments.....38
Risk analysis 39

LIST OF TABLES

Table 1-- System Comparison..... 16
Table 2 - Value Proposition 32
Table 3 - Cost Estimation..... 33
Table 4-Budget and Justification 34
Table 5-Gantt Chart 36
Table 6-Risk Analysis 39

LIST OF FIGURES

Figure 1- system diagram 27
Figure 2- Component diagram 27
Figure 3-Work Breakdown Structure 35

LIST OF ABBREVIATIONS

Abbreviation	Word/Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
GPS	Global Positioning System
RGB	Red, Green, Blue (image color space)
NDVI	Normalized Difference Vegetation Index
ResNet	Residual Neural Network
MobileNet	Lightweight CNN architecture for mobile deployment
DSR	Design Science Research
SDGs	Sustainable Development Goals
UI	User Interface
GIS	Geographic Information System
ML	Machine Learning
DL	Deep Learning
mAP	Mean Average Precision

LIST OF APPENDICES

1. Survey instruments
2. Risk analysis

1.INTRODUCTION

1.1 Background and Context

Rice is one of the most important food crops in the world, serving as the primary staple for more than three billion people across Asia. In Sri Lanka, rice cultivation plays a central role in national food security, rural livelihoods, and cultural identity. The country cultivates approximately 900,000 hectares of paddy fields annually and produces between 2.7 and 3.0 million metric tons of rice each season. Despite this scale of production, actual yields regularly fall below their potential, largely due to damage caused by diseases, pest infestations, and environmental stresses that go undetected until significant crop damage has already occurred [1].

Rice diseases are among the most economically damaging threats to the crop. Rice Blast, caused by the fungus Magnaporthe oryzae, is considered the most destructive rice disease in the world and can cause yield losses of 10 to 50 percent in severely affected regions. Bacterial Leaf Blight, caused by Xanthomonas oryzae pv. oryzae, is another major disease that regularly destroys 20 to 30 percent of a crop when it spreads unchecked. Brown Spot and Sheath Blight are also widely prevalent diseases affecting rice in Sri Lanka and across South and Southeast Asia. According to the Food and Agriculture Organization, rice crop losses due to diseases and pests account for 20 to 40 percent of global production annually, representing economic losses exceeding USD 30 billion each year [2].

Despite the severity of these threats, rice disease monitoring in Sri Lanka and most developing countries continues to rely on traditional manual inspection methods. Farmers or agricultural extension officers conduct periodic visual inspections of rice fields and rely on their experience to identify disease symptoms. This approach has several critical limitations. Manual inspection is time-consuming, spatially incomplete, and inherently reactive — by the time disease symptoms are visible to the human eye, the infection has often already spread across a significant portion of the field. Additionally, the shortage of trained agricultural extension officers in Sri Lanka means that many farms receive infrequent visits, further reducing the effectiveness of traditional monitoring [3].

Advances in artificial intelligence, particularly deep learning and computer vision, have demonstrated strong potential for automated plant disease detection. Convolutional Neural Networks have achieved disease classification accuracy exceeding 95 percent in controlled laboratory settings using crop imagery. However, deploying these systems in real agricultural field conditions introduces significant new challenges, including variable lighting, overlapping canopies, soil and water backgrounds in images, and partial plant occlusions. Most existing research systems were developed and evaluated using isolated leaf images captured indoors, and their performance degrades substantially in real-world field conditions [3].

Existing approaches also tend to focus exclusively on disease classification from static images and neglect several equally important capabilities needed for practical precision agriculture. Early stress prediction — detecting physiological changes in plants before visible disease

symptoms appear can enable preventive treatment that significantly reduces crop losses. Growth stage recognition is important because treatment recommendations and disease vulnerability patterns differ substantially across rice growth stages. Temporal monitoring — tracking crop health changes across multiple field visits over time — provides far richer decision support than any single inspection. And Explainable AI capabilities are needed to make automated detection results understandable and trusted by farmers and agricultural workers who are not AI specialists.

The proposed research addresses these gaps by developing an AI-based crop health monitoring system integrated with an autonomous ground rover platform. The rover captures close-range, high-resolution images of rice plants during field navigation, providing the system with high-quality input imagery under real field conditions. Unlike aerial drone-based systems, which are limited by resolution at altitude and are vulnerable to weather conditions, the ground rover operates at plant level and can capture detailed imagery even under cloud cover or in dense canopy areas.

This research aligns with several United Nations Sustainable Development Goals. It supports SDG 2 (Zero Hunger) by contributing to food security through improved crop protection. It aligns with SDG 9 (Industry, Innovation and Infrastructure) through the development of innovative agricultural technology. It also supports SDG 15 (Life on Land) by enabling targeted, precise pesticide application that reduces environmental impact compared to blanket chemical treatment of rice fields.

The proposed system also aligns with the FoC Research Cluster: Technology Integration and Management, as it combines deep learning, computer vision, and explainable AI techniques with precision agriculture applications to develop a practical and scalable intelligent monitoring platform for rice cultivation.

1.2 Literature Review

Significant research has been conducted on plant disease detection using computer vision and deep learning over the past decade. A review of the most relevant work provides the foundation for the proposed system and highlights the gaps that this research aims to address.

Mohanty et al. [3] conducted a pioneering study on large-scale plant disease recognition using the PlantVillage dataset, which contains over 54,000 images representing 26 diseases across 14 crop species. Using deep CNNs, their system achieved 99.35 percent accuracy under controlled conditions. However, the authors themselves highlighted that performance dropped substantially when the system was applied to images taken in real field conditions rather than isolated leaf photos on clean backgrounds. This gap between laboratory and field performance remains a central challenge that directly motivates the proposed research.

Ferentinos [4] extended this line of work by applying multiple CNN architectures to crop disease detection and demonstrated that transfer learning from pre-trained networks significantly improves classification performance when the labeled domain-specific dataset is small. His system achieved 99.53 percent accuracy on the PlantVillage dataset. The use of transfer learning is particularly relevant to this research, where initial field-specific rice disease data will be limited and will need to be supplemented with augmented and publicly available datasets.

Specifically for rice diseases, Sethy et al. [5] developed a deep CNN-based system trained to detect four major rice diseases including Bacterial Blight, Blast, Brown Spot, and Tungro, achieving 93.8 percent classification accuracy. While their work confirmed the feasibility of CNN-based rice disease classification, it was conducted using isolated leaf images captured under controlled lighting conditions. No spatial mapping, growth stage analysis, or temporal monitoring capabilities were included, and the system was not designed for deployment on mobile agricultural platforms.

Lu et al. [6] proposed a multi-task deep learning approach that simultaneously performed rice disease detection and growth stage estimation using field images. Their framework achieved 89.7 percent disease detection accuracy and 86.4 percent growth stage classification accuracy. The multi-task learning design is directly relevant to the proposed system, which also needs to perform both disease detection and growth stage recognition. However, their work lacked explainability features and did not include temporal analysis or rover integration.

For early stress detection, Mahlein et al. [7] demonstrated that hyperspectral imaging can identify disease-related physiological changes in plants 4 to 7 days before visible symptoms appear. While full hyperspectral camera systems are prohibitively expensive for field rover deployment, their findings confirmed that early spectral and colour changes precede visible symptoms and can therefore be targeted through advanced image analysis of standard RGB imagery. This supports the development of the early stress prediction module in the proposed system.

Fuentes et al. [8] applied Faster R-CNN object detection networks to tomato disease detection and demonstrated that object detection architectures can localize individual disease lesions

within complex field images rather than simply classifying entire images. Their system achieved strong performance even in greenhouse conditions with variable backgrounds. Adapting object detection approaches to localize disease spots within rice plant images captured by the ground rover is a relevant technique for the proposed system.

Regarding Explainable AI, Saleem et al. [9] conducted a comprehensive review of XAI techniques applied to plant disease detection systems, including Grad-CAM, LIME, and SHAP. Their review concluded that interpretability is not only desirable but essential for the practical adoption of AI-based agricultural tools, since farmers and extension officers need to understand why the system made a particular recommendation before they are willing to act on it. This finding directly motivated the inclusion of Grad-CAM visualization in the proposed system.

Zheng et al. [10] evaluated temporal crop health monitoring using sequential time-series field imagery and deep learning, demonstrating that tracking disease progression across multiple observations enables significantly more accurate risk assessment than single-image analysis. Their work showed that temporal trends in disease probability are valuable predictors of crop health deterioration, supporting the temporal monitoring module planned in this research.

Research by Ramesh et al. [11] confirmed that machine learning approaches applied to rice plant imagery can effectively distinguish between healthy plants and multiple disease classes. Their work also emphasized the importance of using field-collected imagery rather than laboratory images to ensure practical system performance.

Finally, Selvaraju et al. [12] introduced Gradient-weighted Class Activation Mapping (Grad-CAM), which uses gradients flowing into the final convolutional layer of a CNN to produce a coarse localization map highlighting the image regions most important to a particular prediction. Their technique is now widely adopted in medical imaging and agricultural AI applications where visual explanation of model decisions is required.

1.3 Research Gap

Even though significant research has been conducted on plant disease detection using computer vision and deep learning, several important gaps remain that the proposed system aims to address.

1. Limited performance in real field conditions: Most existing rice disease detection systems were trained and evaluated using isolated leaf images captured in controlled laboratory environments. These systems achieve high accuracy under clean, consistent lighting with single leaves on plain backgrounds but fail when applied to complex rice field imagery where variable illumination, soil and water backgrounds, overlapping canopies, and partial plant occlusions are common [3]. No existing system has been specifically designed and validated for deployment on an autonomous ground rover operating in real paddy fields.

2. Absence of early stress detection: Most existing systems only classify visible disease symptoms after they have already manifested. They do not attempt to detect the subtle colour and texture changes that occur in rice plants several days before visible lesions appear. Early intervention is far more effective than reactive treatment, but no existing system addresses pre-symptomatic stress detection in rice [7].

3. Lack of integrated multi-capability frameworks: Current systems typically focus on a single task such as disease classification and do not combine disease detection, growth stage recognition, temporal monitoring, and explainable AI within a single integrated platform. A comprehensive crop health intelligence system needs all these capabilities working together to support real agricultural decision-making [6].

4. No explainability for farmer trust: The vast majority of existing deep learning models for rice disease detection are black-box systems that provide classification results without any explanation of why a particular diagnosis was made. As Saleem et al. [9] demonstrated, the lack of interpretability is a major barrier to practical adoption among farmers and agricultural officers who need to understand and trust the system before acting on its recommendations.

Why is this research needed:

This study aims to fill these gaps by developing an AI-based rice crop health monitoring framework that:

- Operates on real field imagery captured by a ground rover, addressing the laboratory-to-field performance gap

- Detects early physiological stress in rice plants before visible disease symptoms appear
- Combines disease detection, growth stage recognition, temporal monitoring, and Grad-CAM explainability within a single integrated system
- Generates GPS-tagged crop health intelligence maps that feed into a broader precision agriculture decision support platform

By integrating these capabilities, the proposed system offers a new, practical approach that no existing system currently provides, making rice crop health monitoring more accurate, transparent, and actionable for farmers, extension officers, and agricultural decision-makers.

Feature	**[3]**	**[5]**	**[6]**	**[8]**	**Existing Drone Systems**	**Proposed System**
Disease Detection	✓	✓	✓	✓ (tomatoes)	✓ Limited	✓
Field / Real Conditions	×	×	Limited	✓ Greenhouse	✓	✓
Early Stress Prediction	×	×	×	×	×	✓
Growth Stage Recognition	×	×	✓ Limited	×	×	✓
Temporal Monitoring	×	×	×	×	×	✓
Explainable AI (Grad-CAM)	×	×	×	×	×	✓
Ground Rover Integration	×	×	×	×	×	✓
Crop Health Intelligence Maps	×	×	×	×	Limited	✓

Table 1-- System Comparison

2. OBJECTIVE

To design, develop, and evaluate an AI-based computer vision system that analyzes ground-level rice plant imagery captured by an autonomous ground rover to detect crop diseases, predict early physiological stress conditions before visible symptoms appear, recognize rice growth stages, perform temporal health monitoring across the cultivation season, and generate GPS-tagged crop health intelligence maps that support precision agriculture decision-making for rice cultivation.

The system will be designed, implemented, and evaluated within the research period, aiming to achieve at least 90 percent accuracy in rice disease classification and demonstrate measurable early stress detection and growth stage recognition capabilities validated against expert agricultural ground truth during the pilot evaluation phase.

2.2 Specific Objectives

1. Rice Disease Detection and Classification

Develop and fine-tune a deep learning model using transfer learning on pre-trained ResNet-50 and MobileNetV3 architectures to classify rice plant images captured by the ground rover into seven categories: Rice Blast, Bacterial Leaf Blight, Brown Spot, Sheath Blight, Tungro, Leaf Scald, and Healthy. The model will be trained on a combined dataset drawn from the PlantVillage dataset, RiceLeafDB, the Paddy Doctor dataset, and primary field images collected from Sri Lankan rice farms across multiple growing seasons and environmental conditions.

The disease detection module will target a minimum classification accuracy of 90 percent and a macro-averaged F1 score of at least 0.88 on the held-out test dataset, ensuring reliable detection performance across all seven classes including less commonly represented disease types.

2. Early Stress Prediction Before Visible Symptoms

Develop an early stress detection module that identifies subtle colour distribution and surface texture changes in rice plant images that indicate early physiological stress before visible disease lesions become apparent to human observers. The module will extract spectral indices from RGB image channels — including the Excess Green Index, the Red-Green ratio, and Local Binary Pattern texture descriptors — as discriminating features, and train a Support Vector Machine classifier on image samples annotated by agricultural experts as pre-symptomatic or healthy.

The early stress prediction module will aim to detect stress conditions at least three to five days before macroscopic symptoms appear, with a detection sensitivity target greater than 80 percent on the held-out evaluation set. This capability enables preventive treatment decisions that are not possible with current reactive manual inspection approaches.

3. Rice Growth Stage Recognition

Implement a dedicated multi-class growth stage classifier using MobileNetV3 that accurately identifies the current development stage of rice plants in rover-captured field imagery. The classifier will recognize the four primary rice growth stages defined by the standard rice development taxonomy: the seedling stage covering 0 to 30 days after transplanting, the vegetative stage covering 31 to 60 days, the reproductive stage covering 61 to 90 days, and the maturity stage covering 91 to 120 days after transplanting.

The classifier will leverage morphological and structural features visible in ground-level imagery, including canopy density, leaf colour distribution, plant height proxy, and panicle presence. Growth stage classification output is used by the system to contextualize disease detection results, since the risk profile and visual presentation of specific diseases differ substantially across rice growth stages. The classifier will target a recognition accuracy of at least 88 percent across all four growth stages.

4. Temporal Crop Health Monitoring

Build a temporal monitoring module that maintains a GPS-indexed crop health history database and compares rice plant analysis results from sequential rover field surveys of the same field zones over time. The module will track disease probability trends, detect zones where health is consistently declining across visits, and generate visual progression charts that show how crop health indicators have changed across the cultivation season for each monitored location.

This temporal analysis enables the system to flag deteriorating field zones for priority treatment even when no individual survey result crosses the disease alert threshold on its own, and provides a longitudinal health record that is far more valuable for agricultural decision-making than any single inspection result. The module will integrate with the GPS positioning data provided by Component 1 of the integrated system to ensure accurate location indexing of all health records.

5. Explainable AI Using Grad-CAM

Incorporate Gradient-weighted Class Activation Mapping (Grad-CAM) into the ResNet-50 disease detection model to generate spatial attention heatmap overlays for each disease-positive classification result. The heatmaps highlight the specific leaf regions and plant features that most strongly influenced the model's classification decision and are overlaid on the original

input image using a colour gradient that moves from blue for low-influence regions to red for the highest-influence regions.

These visual explanations are included alongside the classification label and confidence score in all crop health reports produced by the system. The Grad-CAM component makes the AI decision transparent and verifiable for agricultural extension officers and farmers who have no machine learning background, directly addressing one of the primary barriers to practical adoption of AI-based disease detection systems identified in the literature.

6. Crop Health Intelligence Map Generation

Generate GPS-tagged spatial crop health intelligence maps at the conclusion of each rover field survey by aggregating all disease detection results, early stress flags, growth stage classifications, and temporal trend indicators across all image capture locations into a unified spatial health grid. The maps will visually represent the distribution and severity of crop health conditions across the monitored rice field, enabling extension officers and farmers to immediately identify the zones requiring priority attention and intervention.

The health intelligence maps and their underlying structured data will be output in a format specifically designed to be compatible with the data input requirements of Component 3 (Field Zone Mapping and Seeding Planning) and Component 4 (Predictive Soil Intelligence) of the integrated Smart Drone Compatible Rice Cultivation and Management System, ensuring seamless data flow across the full precision agriculture platform.

3. METHODOLOGY

Research Approach

This research proposes the development of an AI-based computer vision system designed to monitor rice crop health using imagery captured by an autonomous ground rover during field navigation. Rice diseases such as Blast, Bacterial Leaf Blight, and Brown Spot cause substantial crop losses each year in Sri Lanka and across Asia, yet current monitoring approaches depend almost entirely on manual, periodic visual inspection by farmers or extension officers. This approach is reactive rather than preventive — disease is typically noticed only after symptoms have already spread across a significant portion of the field. The shortage of trained agricultural extension officers in Sri Lanka further reduces how often and how thoroughly fields can be monitored.

To overcome these limitations, the proposed system introduces a deep learning pipeline that combines rice disease detection, early stress prediction, growth stage recognition, temporal crop monitoring, and Explainable AI within a single integrated platform. The system processes ground-level rice plant images captured by the rover and produces structured crop health intelligence outputs, including GPS-tagged health maps and Grad-CAM visual explanation overlays, that support agricultural decision-making across the full cultivation season.

This research follows a Design Science Research (DSR) methodology, which is appropriate because the primary goal is to design and develop a novel technological artifact — the AI crop health monitoring system — and rigorously evaluate its effectiveness against defined performance criteria. The study also incorporates an experimental evaluation component in which the system's disease classification accuracy, early stress detection sensitivity, and growth stage recognition performance are measured and compared against ground truth labels established through expert agricultural validation. This combination of artifact development and empirical evaluation ensures that the system is both technically sound and practically relevant.

Technical Methods and Algorithms

The system collects and analyzes rice plant imagery through a multi-stage AI pipeline. Each stage targets a specific analytical objective and uses appropriate machine learning techniques to produce the required output.

Dataset Preparation and Preprocessing

Two complementary data sources will be combined to build the training dataset. The primary dataset consists of ground-level rice plant images collected from rice paddies in Sri Lanka using the project rover's camera. These images will cover multiple rice varieties, disease states, and all four growth stages, captured under varied real-world lighting and environmental conditions. The secondary dataset draws from three publicly available repositories: the PlantVillage dataset, which contains over 54,000 labeled crop disease images including rice; the

RiceLeafDB dataset, which specifically covers rice leaf diseases; and the Paddy Doctor dataset, which includes over 10,000 paddy field images across ten disease and pest categories.

All collected images are preprocessed before being fed into the AI models. Preprocessing steps include:

- Resizing all images to 224 by 224 pixels to match the standard input size of ResNet-50 and MobileNetV3 architectures
- Normalizing pixel values to the 0 to 1 range to stabilize model training
- Applying data augmentation techniques including random horizontal and vertical flipping, rotation of up to 30 degrees, brightness and contrast variation, Gaussian noise addition, and random cropping

Data augmentation is applied to simulate the natural variability of real paddy field conditions and reduce the risk of model overfitting to the specific characteristics of the training images.

Disease Detection Module

The core disease detection module uses transfer learning to fine-tune two pre-trained Convolutional Neural Network architectures on the combined rice disease dataset. ResNet-50 is selected as the primary model because its deep residual architecture and skip connections allow it to learn complex rice disease visual patterns without suffering from the vanishing gradient problem that affects simpler deep networks. MobileNetV3 is trained as a secondary lightweight model because its depthwise separable convolutions significantly reduce the number of parameters, making it suitable for potential deployment on embedded hardware directly on the rover.

For both models, only the final classification layers are replaced with custom softmax output layers corresponding to the seven target classes: Rice Blast, Bacterial Leaf Blight, Brown Spot, Sheath Blight, Tungro, Leaf Scald, and Healthy. The lower convolutional layers are initially frozen to preserve the generalizable low-level feature representations learned during pre-training on ImageNet. The upper layers are then progressively unfrozen and fine-tuned using a low learning rate to adapt the models to rice plant imagery. The training process combines categorical cross-entropy loss with the Adam optimizer and early stopping based on validation loss to prevent overfitting.

Early Stress Detection Module

Early stress detection is implemented as a two-stage analytical process designed to identify physiological changes in rice plants before visible disease lesions appear. Research has shown that disease-related cellular changes in leaves alter their optical properties — specifically their colour distribution and surface texture — several days before macroscopic symptoms become visible to the human eye.

In the first stage, spectral indices and texture features are extracted from the RGB channels of each rover-captured image. These include:

- Excess Green Index (ExG), calculated as 2G minus R minus B, which amplifies green vegetation signals and is sensitive to early chlorophyll reduction
- Red-Green ratio, which increases as leaves begin to yellow or develop early necrotic patches
- Colour histogram features capturing the distribution of hue, saturation, and value across the leaf region
- Local Binary Pattern (LBP) texture descriptors that capture subtle surface irregularities not visible in colour alone

In the second stage, these extracted features are used to train a Support Vector Machine (SVM) classifier on image samples annotated by agricultural experts as either pre-symptomatic stressed or healthy. A radial basis function kernel is used to handle the non-linear feature boundaries between stress classes. A sliding window temporal comparison additionally flags field zones where colour index values show a consistent worsening trend across two or more consecutive rover visits, even if no individual image crosses the stress detection threshold on its own.

Growth Stage Recognition

A dedicated MobileNetV3 multi-class classifier is trained specifically for rice growth stage identification. The four target growth stages follow the standard rice development taxonomy: seedling stage covering 0 to 30 days after transplanting, vegetative stage covering 31 to 60 days, reproductive stage covering 61 to 90 days, and maturity stage covering 91 to 120 days after transplanting.

The classifier relies on several morphological and structural features that change distinctly across growth stages and are visible in ground-level rover imagery. These include:

- Canopy density and leaf coverage area, which increases progressively from seedling through vegetative to reproductive stages
- Leaf colour distribution, which shifts from bright yellow-green in seedlings to deep green in the vegetative stage and then to golden-yellow tones at maturity
- Panicle presence, which is the most reliable indicator distinguishing the reproductive and maturity stages from earlier vegetative stages
- Plant height proxy estimated from image perspective geometry and the vertical extent of the plant within the camera frame

Growth stage labels for the primary field training images are assigned based on planting date records provided by farmers combined with direct expert observation during field visits. The growth stage classifier output is used by the broader system to contextualize disease detection results, since the risk and visual appearance of specific diseases differ across rice growth stages.

Temporal Crop Monitoring Module

The temporal monitoring module maintains a GPS-indexed health history database that records disease detection results, stress flags, and growth stage classifications for each field location across multiple rover survey visits. When the rover revisits the same field zones on subsequent surveys, the new analysis results are compared against the stored historical records for those zones.

The comparison uses disease probability trend analysis to identify zones where health is declining across visits, even if no individual survey result exceeds the alert threshold on its own. Structural Similarity Index Measure (SSIM) comparisons between sequentially captured images of the same zone provide an additional indicator of visual change that complements the CNN probability outputs. The module generates progression charts showing how disease probability and stress indicators have changed over time for each monitored field zone, and automatically flags zones showing consistent deterioration for priority attention in treatment planning.

Explainable AI — Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the ResNet-50 disease detection model to generate spatial attention heatmaps for each disease-positive image classification result. Grad-CAM works by computing the gradients of the target disease class prediction score with respect to the feature activation maps in the final convolutional layer of the network. These gradients are used as importance weights to produce a coarse spatial map highlighting which regions of the input image most strongly influenced the classification decision.

The resulting heatmap is upsampled and overlaid on the original input image using a colour gradient that moves from blue for low influence regions through yellow to red for the highest influence regions. Disease-positive detections with their corresponding Grad-CAM overlays are included in crop health reports alongside the classification label and confidence score. This allows agricultural extension officers and farmers to see exactly which part of a rice plant caused the system to diagnose a particular disease, making the AI decision transparent and verifiable without requiring any machine learning knowledge.

Tools and Platforms with Justification

The system is developed using the following tools and platforms, each selected for specific technical reasons relevant to AI-based image analysis and agricultural deployment.

Python 3.10 is used as the primary development language because it is the de facto standard for AI and machine learning development and offers the most comprehensive ecosystem of libraries specifically designed for deep learning and computer vision tasks.

TensorFlow and Keras are used for CNN model development, transfer learning, and training pipeline implementation. The high-level Keras API simplifies the construction and training of ResNet-50 and MobileNetV3 architectures, and TensorFlow provides efficient support for loading pre-trained ImageNet weights and GPU-accelerated model training.

PyTorch is used specifically for the Grad-CAM implementation. PyTorch's dynamic computation graph allows gradient values to be captured from intermediate convolutional layers at inference time, which is required for Grad-CAM heatmap generation. This operation is considerably more straightforward in PyTorch than in TensorFlow's static graph model.

OpenCV is used for all image preprocessing operations including resizing, normalization, histogram equalization, and spectral index computation. OpenCV is optimized for efficient real-time image processing and provides the computer vision operations needed for the early stress detection module's feature extraction pipeline.

scikit-learn is used for the Support Vector Machine classifier in the early stress detection module and for computing evaluation metrics including classification accuracy, precision, recall, F1 score, and confusion matrices across all modules.

Roboflow is used for structured dataset annotation, version management, and augmentation pipeline configuration. Roboflow provides a specialized environment for managing agricultural image datasets with an efficient labeling interface and built-in support for the augmentation strategies needed to prepare the rice disease training dataset.

For development and collaboration, Git is used for version control with the repository hosted on GitHub. The system is developed using Visual Studio Code as the main integrated development environment. Google Colab Pro with GPU acceleration is used for model training to access the computational resources required for deep learning without local hardware constraints.

Data Collection and Ethical Considerations

Data for this research will be collected from two main sources: primary field imagery collected from rice farms and expert consultation with agricultural and plant pathology professionals.

Primary field image data will be collected from rice paddy farms in the Western, North Western, and North Central Provinces of Sri Lanka, which represent the primary rice cultivation zones in the country. Field visits will be scheduled across multiple stages of the rice growing season to ensure that imagery is captured for all four growth stages and under varying disease conditions and environmental settings. The rover camera will be used to capture close-range plant images during each visit, with GPS coordinates recorded for every image capture location.

Expert knowledge will be gathered from plant pathologists and agricultural extension officers at the Department of Agriculture Sri Lanka through structured interviews and dataset labelling sessions. Their expertise will guide the assignment of disease and pre-symptomatic stress labels in the primary dataset and will be used to validate the system's classification outputs during the evaluation phase.

All data collection procedures will follow appropriate ethical research practices. Informed consent will be obtained from all participating farmers before any field imagery is collected on their land, using a formal consent form that explains the research purpose and how the collected data will be used. All images will be anonymized by removing embedded GPS metadata before storage in the research dataset. No personal information about participating farmers or landowners will be collected, stored, or included in any research publication.

Validation Strategy and Performance Metrics

Once the system is developed, it will be evaluated through a structured validation process covering technical accuracy, expert agreement, and practical usability. The combined dataset will be divided into 80 percent training, 10 percent validation, and 10 percent test sets. The test set images will be held completely separate from training and validation data throughout the development process to ensure unbiased performance evaluation.

Technical Validation

The performance of the disease detection CNN, the early stress SVM classifier, and the growth stage classifier will each be evaluated independently against expert-labeled ground truth on the held-out test set. Overall classification accuracy, macro-averaged precision, recall, F1 score, and a full multi-class confusion matrix will be computed for each module to identify which disease classes are most likely to be confused with one another and where further training improvements may be needed.

Expert Validation

A representative sample of classification results along with their corresponding Grad-CAM heatmap overlays will be presented to at least two qualified plant pathologists and two agricultural extension officers for independent review. They will assess whether each disease diagnosis is correct and whether the highlighted plant region in the Grad-CAM overlay corresponds to the actual location of visible disease indicators in the image. This qualitative expert review is particularly important for evaluating the practical reliability and interpretability of the explainability component.

Usability Evaluation

The crop health reports and intelligence maps generated by the system will be reviewed by agricultural extension officers to evaluate whether the output format is clear, actionable, and suitable for use in real field management decisions. Feedback will be collected through structured interviews and used to refine the report design before the final system submission.

Performance Targets

The system aims to achieve the following outcomes:

- Disease classification accuracy of at least 90 percent on the held-out test dataset
- Macro-averaged F1 score of at least 0.88 across all seven disease and healthy classes
- Early stress detection sensitivity greater than 80 percent on pre-symptomatic test samples
- Growth stage recognition accuracy of at least 88 percent across all four rice growth stages
- Disease classification inference time of no more than 500 milliseconds per image on the target processing platform
- Grad-CAM heatmaps assessed as correctly localizing visible disease indicators by at least 75 percent of reviewing agricultural experts

4.HIGH-LEVEL SYSTEM ARCHITECTURE

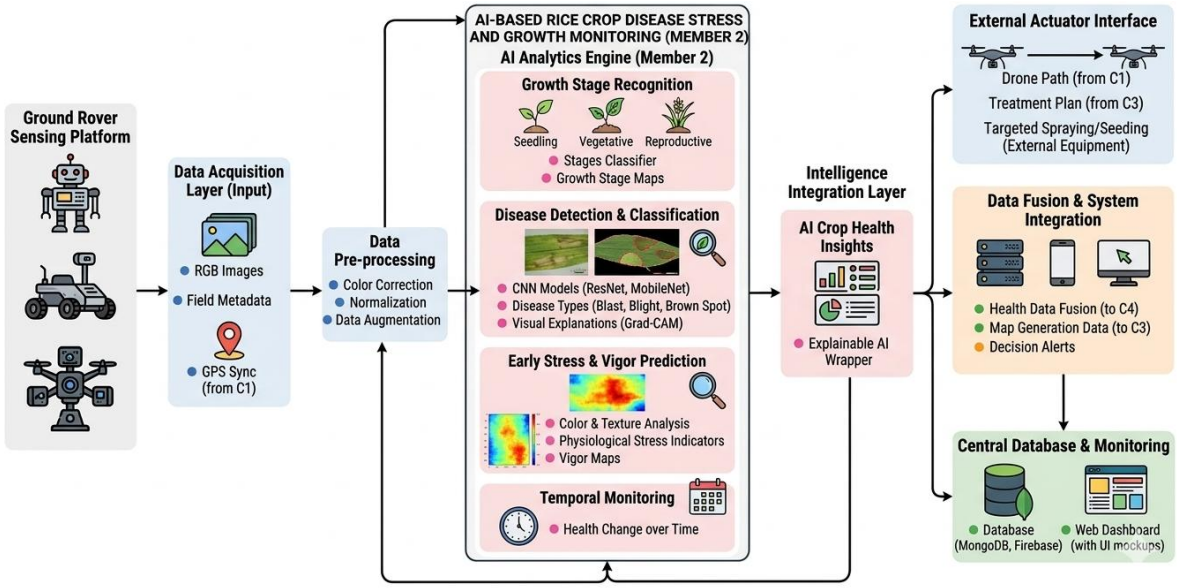


Figure 1- system diagram

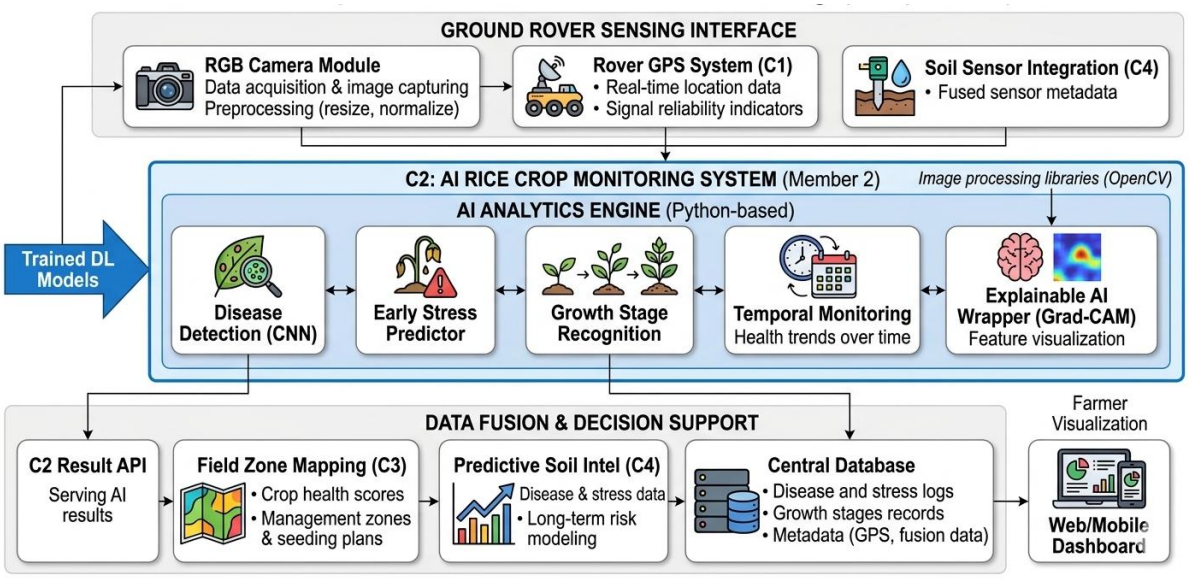


Figure 2- Component diagram

5.USER REQUIREMENTS

Stakeholder identification

The proposed AI crop health monitoring system involves several stakeholders who will interact with the system in different ways.

Rice Farmers (Primary Users):

Rice farmers are the primary beneficiaries of the system. They interact with the system indirectly through the crop health reports and field maps generated after each rover survey. Farmers use disease alerts and treatment recommendations to decide when and where to apply pesticides, fertilizers, or other interventions.

Agricultural Extension Officers (Secondary Users):

Extension officers use the crop health intelligence maps and Grad-CAM visualizations to prioritize field visits, plan district-level disease interventions, and provide informed guidance to farmers based on spatially accurate disease distribution data.

System Operators and Rover Technicians (Tertiary Users):

Operators are responsible for deploying and maintaining the ground rover and ensuring the AI system is functioning correctly. They interact with the system configuration interface and monitor data collection quality.

Component 3 and Component 4 Teams (Internal System Users):

The field zone mapping and soil intelligence components of the integrated system consume the structured data outputs of this component as inputs for their own processing. These are internal system-level stakeholders rather than end-user stakeholders.

.

Functional Requirements

The system must support several core functions to capture, analyze, and report rice crop health information.

1. The system should capture and process rice plant images from the ground rover camera and associate each image with a GPS coordinate and timestamp.
2. The system should classify each rice plant image into one of seven categories: Rice Blast, Bacterial Leaf Blight, Brown Spot, Sheath Blight, Tungro, Leaf Scald, or Healthy.
3. The system should analyze rice plant images for early stress indicators and flag plants showing pre-symptomatic physiological changes before visible lesions appear.
4. The system should classify the rice growth stage of each plant in captured images across the four primary stages: seedling, vegetative, reproductive, and maturity.

5. The system should maintain a GPS-indexed temporal health database and compare current-visit images with previous-visit records to track health trends over time.
6. The system should generate Grad-CAM heatmap visualizations for disease-positive detections, highlighting the plant regions that influenced the classification result.
7. The system should produce GPS-tagged crop health intelligence maps at the conclusion of each rover field survey, aggregating all module outputs into a spatial health grid.
8. The system should output structured data in a format compatible with the data input requirements of Component 3 and Component 4 of the integrated precision agriculture system.
9. The system should generate crop health summary reports that include disease type, detection confidence, GPS location, growth stage, and timestamp for each flagged plant location.
10. The system should support model retraining with new field data through a defined pipeline without requiring full system re-architecture.

Non-Functional Requirements

Non-functional requirements describe how well the system performs and the quality attributes it must maintain.

Performance:

The system should complete disease classification inference within 500 milliseconds per image to support near-real-time monitoring during rover navigation.

Accuracy:

Disease classification accuracy must exceed 90 percent on the held-out test dataset to ensure reliable detection results for agricultural decision-making.

Reliability:

The system must maintain stable and consistent performance across varying field lighting conditions including overcast skies, partial shade from canopy, and bright sunlight.

Usability:

Crop health maps and reports must present information in a visual format that can be understood and acted upon by agricultural extension officers without technical training in machine learning or computer vision.

Security:

All image data and health intelligence outputs must be transmitted via encrypted channels. Access to health data must be restricted to authorized agricultural stakeholders.

Scalability:

The system architecture must support the addition of new disease classes and the incorporation of additional rover survey data without requiring full system redesign.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system is rice farmers, agricultural cooperatives, and government agricultural extension services across Sri Lanka and Southeast Asia. The system is particularly relevant for medium to large scale rice cultivation operations covering more than five acres where the cost of manual disease monitoring and undetected crop loss is significant.

The main customers who would adopt the system include:

Rice farmers and agricultural cooperatives:

Farmers and cooperative organizations managing paddy cultivation who experience recurring crop losses due to late disease detection and wish to reduce pesticide expenditure through targeted, need-based treatment.

Government agricultural extension services:

The Department of Agriculture Sri Lanka and equivalent bodies in other rice-producing nations who need scalable, automated crop health surveillance tools to support national food security programs.

Agri-tech companies and precision farming solution providers:

Technology companies building integrated precision agriculture platforms who can incorporate this AI component as part of a broader smart farming solution.

Crop insurance companies:

Insurance providers who could use verified crop health monitoring data to assess risk and determine payouts more accurately.

The rice production market in Sri Lanka alone involves over 900,000 hectares of paddy land. Globally, over 165 million hectares of rice are cultivated annually, representing a substantial addressable market for precision agriculture health monitoring tools.

Value Proposition

Value Element	Description
Early Disease Detection	Detects diseases 3–5 days before visible symptoms appear, enabling preventive treatment that reduces crop losses by 15–25%
Reduced Input Costs	Targeted pesticide application based on precise disease maps reduces chemical input costs by 20–35% compared to blanket treatment
Yield Protection	Early intervention protects 10–15% additional yield per season compared to reactive manual monitoring approaches

Trusted Explainability	Grad-CAM visual explanations show exactly which plant features triggered each disease alert, building farmer and extension officer trust
Integrated Intelligence	Crop health data feeds directly into zone mapping and soil analysis for comprehensive whole-farm decision support
Accessible Reporting	Visual health maps and summary reports are designed for use by agricultural officers without any AI or technical background
Continuous Monitoring	Temporal tracking across multiple rover visits provides a full-season health record rather than isolated inspection snapshots

Table 2 - Value Proposition

Revenue Model

A hybrid B2B and B2G (Business-to-Government) revenue model is proposed:

1. Government and Cooperative Licensing

Annual software licensing agreements for national agricultural extension services and regional farming cooperatives. Multi-farm, district-level access packages.

2. SaaS Platform Subscription

Cloud-based access to the AI platform for individual farmers or extension services on a per-hectare-per-season subscription model. Tiered plans based on farm size and monitoring frequency.

3. Data Analytics Services

Aggregated and anonymized crop health trend data made available to agrochemical companies, agricultural research institutions, and government policy planners for market intelligence and planning purposes.

4. Integration Licensing

Licensing the AI component to agri-tech companies who wish to integrate it as part of their own precision farming solutions or autonomous rover systems.

Cost Estimation

Cost Category	Estimated Cost (LKR)	Details
Field Data Collection	Rs. 30,000 – 50,000	Transportation to rice farms across multiple provinces, multiple field visits across growing seasons

Dataset Annotation	Rs. 15,000 – 25,000	Agricultural expert sessions for disease labelling and pre-symptomatic stress annotation using Roboflow
Cloud GPU Compute	Rs. 20,000 – 30,000	Google Colab Pro+ or AWS GPU instances for CNN training and hyperparameter optimization
Expert Consultation	Rs. 10,000 – 15,000	Compensation for plant pathologists and extension officers for system validation and result review
Testing and Piloting	Rs. 10,000 – 15,000	Field testing across multiple farm locations, validation data collection
Printing and Documentation	Rs. 2,000	Printing proposals, consent forms, and field survey sheets
TOTAL INITIAL COSTS	Rs. 87,000 – 137,000	

Table 3 - Cost Estimation

Pricing Strategy

A penetration pricing strategy will be used during the initial deployment phase to attract early adopters in government extension services and farming cooperatives. Pricing will be set at LKR 20,000–35,000 per farm per cultivation season for farms up to 50 acres, which represents less than 10 percent of the average seasonal pesticide expenditure for a farm of that scale and is therefore cost-effective even at current performance projections. Government and cooperative bulk contracts will receive discounted institutional rates. Pricing will be reviewed and adjusted as the system scales and additional features are added.

Competitive Advantage

The proposed system offers several advantages compared to existing commercial agricultural AI tools. The ground rover integration provides higher-resolution, plant-level imagery than aerial drone-based systems and is less affected by weather or canopy height. The Grad-CAM explainability component builds farmer trust through transparent, visually interpretable results that no competing system currently provides. The integrated multi-capability framework — combining disease detection, early stress prediction, growth stage recognition, temporal monitoring, and spatial health mapping within a single system — is more comprehensive than any standalone disease alert application currently available. The system is also specifically calibrated for Sri Lankan and South Asian rice varieties and disease profiles, giving it an advantage over generic global solutions that were not trained on locally relevant data.

Intellectual Property Considerations

The intellectual property of the proposed system includes the custom CNN fine-tuning pipeline, the early stress detection algorithm combining spectral index features with

SVM classification, the temporal health monitoring module, and the integrated crop health intelligence map generation system. Copyright protection will be sought for the software codebase. The novel combination of ground rover imagery, temporal analysis, and Grad-CAM explainability for rice disease monitoring may qualify for patent consideration upon commercialization.

7.BUDGET AND JUSTIFICATION

Cost Category	Estimated Cost (LKR)	JUSTIFICATION
Field Data Collection Travel	Rs. 15,000	Transportation costs for multiple visits to rice farms in the North Central and North Western Provinces of Sri Lanka to collect primary field imagery across different growth stages and disease conditions.
Agricultural Expert Consultation	Rs. 12,000	Structured sessions with plant pathologists and agricultural extension officers at the Department of Agriculture Sri Lanka for dataset disease labelling validation and system output evaluation.
Cloud GPU Compute	Rs. 20,000	Google Colab Pro+ subscription and AWS EC2 GPU instance hours for ResNet-50 and MobileNetV3 transfer learning training and hyperparameter search runs.
Dataset Annotation (Roboflow)	Rs. 8,000	Roboflow professional plan for structured image annotation, dataset version management, and augmentation pipeline management.
Printing and Documentation	Rs. 2,000	Printing of research proposal, ethics consent forms, field data collection sheets, and interview guides.
Contingency	Rs. 3,000	Reserve for unforeseen expenses during field data collection or software development phases.
TOTAL	Rs. 60,000	

Table 4-Budget and Justification

8.WORK BREAKDOWN STRUCTURE (WBS)

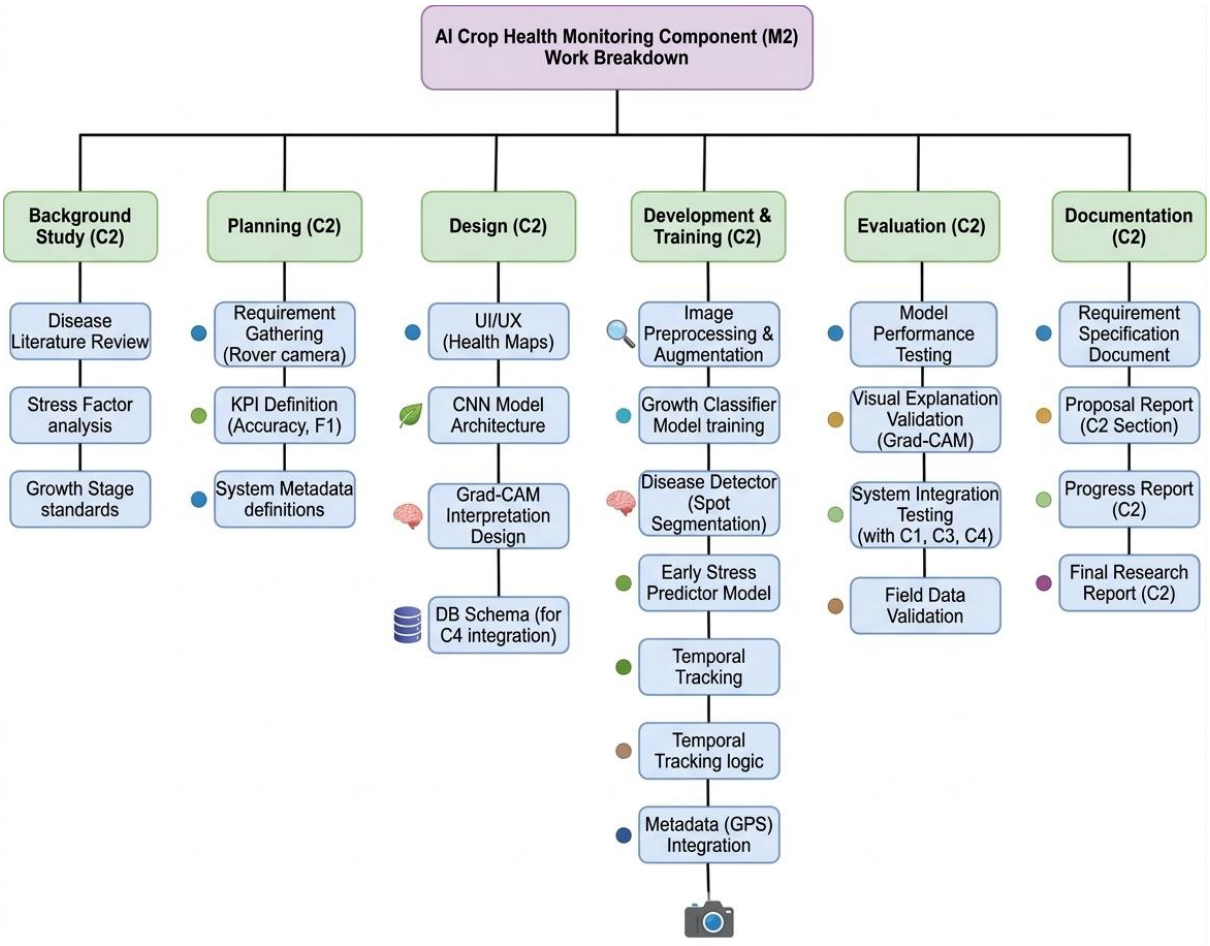


Figure 3-Work Breakdown Structure

9. GANTT CHART

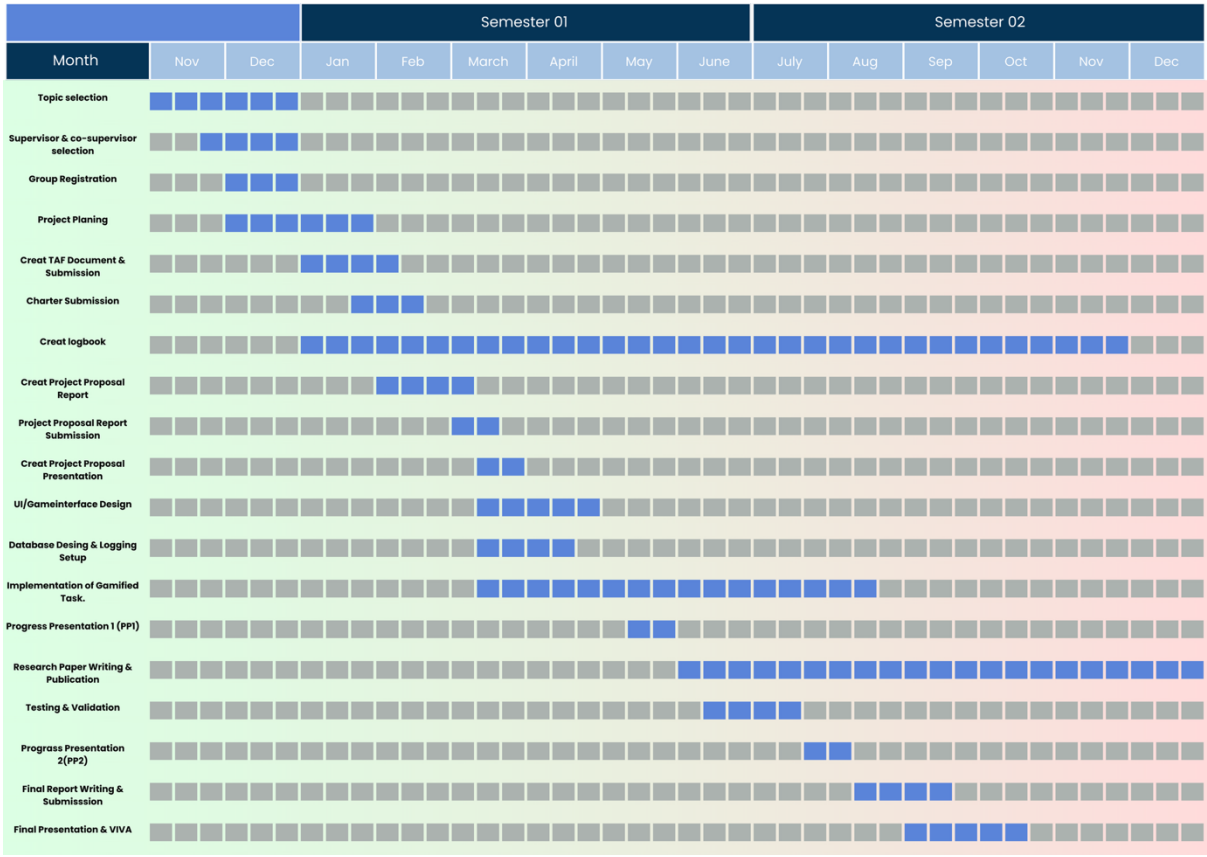


Table 5-Gantt Chart

10. REFERENCES

[1] F. a. A. O. o. t. U. Nations, "The State of Food and Agriculture 2022," FAO, Rome, Italy, 2022.

[2] S. W. L. P. S. J. E. P. M. N. N. A. Savary, "The global burden of pathogens and pests on major food crops," *Nature Ecology & Evolution,* vol. 3, no. 3, p. 430–439, 2019.

[3] S. P. H. D. P. S. M. Mohanty, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science,* vol. 7, p. 1–10, 2016.

[4] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture,* vol. 145, p. 311–318, 2018.

[5] P. K. B. S. K. R. A. K. B. S. Sethy, "Deep feature based rice leaf disease identification using support vector machine," *Computers and Electronics in Agriculture,* vol. 175, 2020.

[6] Y. Y. S. Z. N. L. Y. Z. Y. Lu, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing,* vol. 267, p. 378–384, 2017.

[7] A. K. Mahlein, "Plant disease detection by imaging sensors — parallels and specific demands for precision agriculture and plant phenotyping," *Plant Disease,* vol. 100, no. 2, p. 241–251, 2016.

[8] A. Y. S. K. S. C. P. D. S. Fuentes, "A robust deep-learning-based detector for real-time tomato plant diseases and pests recognition," *Sensors,* vol. 17, no. 9, 2017.

[9] M. H. P. J. A. K. M. Saleem, "Plant disease detection and classification by deep learning," *Plants,* vol. 8, no. 11, 2019.

[10] X. L. X. S. T. C. Y. Zheng, "Temporal analysis of crop health using sequential field imagery and deep learning," *Computers and Electronics in Agriculture,* vol. 195, 2022.

[11] S. H. R. N. M. P. R. S. N. P. V. S. Ramesh, "Plant disease detection using machine learning," Bangalore, India, 2018.

[12] R. R. C. M. D. A. V. R. P. D. B. D. Selvaraju, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision,* vol. 128, no. 2, p. 336–359, 2020.

11.APPENDICES

Survey instruments

AI Based Rice Crop Disease, Stress and Growth Monitoring

Introduction

AI Based Rice Crop Disease, Stress and Growth Monitoring is a component of a fourth-year undergraduate research project in Information Technology at SLIIT. The research focuses on designing an AI-based computer vision system that uses imagery from an autonomous ground rover to detect rice crop diseases, predict early stress conditions, and identify growth stages.

The purpose of this questionnaire is to gather professional insights from plant pathologists, agricultural extension officers, and rice cultivation specialists about current disease identification practices, common disease patterns, and practical field monitoring challenges in Sri Lanka.

The information collected will be used only for academic research and system development purposes. It will help ensure that the AI system is designed to address real agricultural needs. All responses will remain confidential and will be used only to improve the accuracy, relevance, and practical utility of the proposed system.

Questions

1. What are the most economically significant rice diseases affecting paddy cultivation in Sri Lanka, and in what order of priority would you rank them for early detection?
2. At what stage of disease progression do farmers or extension officers typically first notice symptoms during a standard field inspection? How much of the crop is usually affected by this point?
3. Are there any colour changes, texture differences, or leaf morphological indicators in rice plants that experienced agronomists can detect before full visible lesions appear? Can you describe these early indicators?
4. How do disease symptom appearances differ across the four main rice growth stages — seedling, vegetative, reproductive, and maturity?
5. What environmental conditions, such as temperature, humidity, or water logging, most strongly correlate with disease outbreaks in local Sri Lankan conditions?
6. What format of disease detection output — for example, disease type, severity estimate, GPS location, or visual heatmap — would be most useful for guiding field treatment decisions?
7. Would a visual explanation showing the specific leaf region that triggered a disease detection result increase your confidence in acting on the system's recommendation?
8. What false positive rate, where the system detects disease where none is present, would be acceptable for a practical field deployment of an AI disease monitoring system?
9. Are there any specific rice varieties widely grown in Sri Lanka that present unusual or atypical disease symptoms that the system should be aware of?

10. What additional information or features would make an AI-based crop health monitoring report most useful in your professional practice?

Risk analysis

A risk matrix is used to evaluate potential technical and operational risks that may occur during the development and use of the AI-based rice crop health monitoring system, including the disease detection module, early stress prediction module, growth stage classifier, and temporal monitoring system.

Risk Factor	Impact	Probability	Mitigation Strategy
Insufficient primary field image data for model training	High	Medium	Supplement with augmented data and transfer learning from PlantVillage, RiceLeafDB, and Paddy Doctor public datasets. Schedule multiple field collection visits across the growing season.
Poor model accuracy under complex field lighting and canopy conditions	High	Medium	Include training images captured under diverse lighting conditions. Apply histogram equalization and adaptive brightness preprocessing to reduce sensitivity to illumination variation.
Early stress detection misclassification (false positives flagging healthy plants)	High	Medium	Validate pre-symptomatic stress labels with at least two independent agricultural experts. Set conservative detection thresholds and flag uncertain predictions for human review rather than generating automatic alerts.
Dataset labelling errors from non-expert annotation	High	Medium	Engage qualified plant pathologists for cross-validation of all disease and stress class labels. Implement an inter-annotator agreement check before finalizing the training dataset.
Overfitting to training data patterns	High	Medium	Apply L2 regularization, dropout layers, and extensive data augmentation. Monitor validation loss closely during training and apply early stopping when validation performance degrades.
Integration failures with Component 3 and 4 data feeds	Medium	Low	Define clear API specifications and data output formats for inter-component communication early in the development phase. Conduct integration testing with Component 3 and 4 teams before final submission.
Rover camera image quality variation affecting detection accuracy	Medium	Low	Define minimum camera specification requirements. Implement an image quality screening step before feeding images into the AI pipeline, rejecting images that fall below a minimum sharpness and resolution threshold.
Data privacy concerns with field image collection	Low	Low	Remove GPS metadata from all images before storage. Obtain signed informed consent from all participating farmers. Follow SLIIT ethics guidelines throughout data collection.

Table 6-Risk Analysis

Smart Agri-Drone: Intelligent Variable-Rate Fertilizer and Seeding Mission Control

R26-IT-075

Project Proposal Report

S G M B Subasinghe - IT22276032

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
S G M B Subasinghe	IT22276032	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Samantha Rajapaksha	14/03/2026	

ABSTRACT

The cultivation of rice is one of the core activities in agricultural practices in Sri Lanka as well as in the rest of Asia where rice is cultivated to serve the livelihoods of millions of farming households. Rice fields should be properly managed by making right decisions in time about seeding of the fields, application of fertilizers and special attention to problem areas. The conventional field management methods rely a lot on homogenous application programs that do not factor in the spatial differences in crop health and soil conditions and which result in ineffective utilization of the resources and low crop produce.

This study suggests a spatial analysis and planning platform that transforms the data of crop health intelligence and soil measurements, which the autonomous ground rover gathers, into organized rice field management areas and executable seeds planting and treatment plans. The data that is fed into the system is crop health indicators based on the AI based crop monitoring component and soil condition predictions based on the soil intelligence component of the integrated rover platform. The system splits the rice field into separate management zones with the help of clustering algorithms and the method of geospatial analysis with references to vegetation health, soil quality, and historical performance data.

All the identified areas are assigned based on their agronomic status, and the system generates the guides to the agronomic changes in seeding density, fertilizer application rates, and the specific agronomic interventions, including the need to introduce additional irrigation or to take care of crops. A mission planning map is generated to plan the field work and a visualization dashboard display zone boundaries, soil status and the suggested action in user-friendly format to the farmers and agricultural practitioners.

The python language is used to implement this system together with geospatial analysis libraries, K-means and DBSCAN clustering algorithms, and visualization tools of GIS. A validation is done on simulated agricultural data as well as expert agronomic data.

The anticipated results are proper classification of field zoning, sound seeding density advice that is in harmony with the agronomic conditions, and a workable spatial planning instrument that will minimize the input expenses and enhance the efficiency of the rice production.

Keywords Rice Field Zone Mapping, Precision Agriculture, Seeding Recommendation, Treatment Planning, Spatial Analysis, K-means Clustering, GIS, Autonomous Ground Rover.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of my fourth-year research project.

First and foremost, I would like to thank my Research Project Supervisor for their continuous guidance, encouragement, and valuable support throughout the entire research process. Their advice, expertise, and constructive feedback played a significant role in the development of this proposal.

I would also like to extend my sincere appreciation to my Co-Supervisor for the valuable insights and guidance provided during the planning and design stages of this research component.

My special thanks go to the agricultural experts and agronomists who generously shared their professional knowledge regarding rice cultivation practices, field management strategies, and precision agriculture techniques. Their insights greatly contributed to shaping the research direction.

I would also like to thank my group members - the team behind the Smart Rice Field Intelligence System - for their teamwork, cooperation, and dedication throughout this integrated project. The collaborative design of the overall rover platform and data flow architecture was essential in defining the scope and inputs of this component.

Finally, I would like to express my heartfelt gratitude to my family and friends for their encouragement, patience, and continuous support throughout the entire research journey.

TABLE OF CONTENT

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 6

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

LIST OF APPENDICES 9

1.INTRODUCTION 10

1.1 Background and Context 10

1.2 Literature Review 11

1.3 Research Gap 13

2. OBJECTIVE 14

2.1 Main Objectives 14

2.2 Specific Objectives 1. Field Zone Segmentation and Classification 15

2. Seeding Recommendation Generation 15

3. Treatment Planning and Intervention Scheduling 15

4. Mission Planning Map Generation 16

5. Visualization Dashboard Development 16

3. METHODOLOGY 17

Research Approach 17

Technical Methods and Algorithms 17

Data Input and Preprocessing 17

Field Zone Segmentation Algorithm 17

Zone Classification Framework 18

Rule-Based Seeding Recommendation Engine 18

Treatment Planning Logic 18

Tools and Platforms with Justification 18

Data Collection and Ethical Considerations 19

Validation Strategy and Performance Metrics 20

4.HIGH-LEVEL SYSTEM ARCHITECTURE 20

5.USER REQUIREMENTS 21

Stakeholder identification 21

Functional Requirements 22

Non-Functional Requirements 22

6. COMMERCIALIZATION PLAN 23

Target Market and Customer Persona 23

Value Proposition 24
Revenue Model 24
Cost Estimation 25
Competitive Advantage ..... **Error! Bookmark not defined.**
Pricing Strategy 25
Competitive Advantage 25
Intellectual Property Considerations 25
7.BUDGET AND JUSTIFICATION 26
8.WORK BREAKDOWN STRUCTURE (WBS) 27
9. GANTT CHART 28
10 .REFERENCES LIST 29
11.APPENDICES 29
Survey instruments 29
Risk analysis 30

LIST OF TABLES

Table 1-System Comparison 14
Table 2-Value Proposition 24
Table 3-Cost Estimation 25
Table 4-Budget and Justification 26
Table 6-Risk Analysis 30

LIST OF FIGURES

Figure 1-system diagram 21
Figure 2-Component Diagram 21
Figure 3-Budget and Justification 27
Figure 4-Gantt Chart 28

LIST OF ABBREVIATIONS

Abbreviation	Word/Description
GIS	Geographic Information System
GPS	Global Positioning System
NDVI	Normalized Difference Vegetation Index
SDGs	Sustainable Development Goals
WBS	Work Breakdown Structure
DSR	Design Science Research
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
FAO	Food and Agriculture Organization
CSV	Comma-Separated Values
UI	User Interface
IDE	Integrated Development Environment
IP	Intellectual Property

LIST OF APPENDICES

1.survey Instrument……..30
2.Risk Analysis…….31

1.INTRODUCTION

1.1 Background and Context

Rice (Oryza sativa) is among the most important staple crops in the world as it forms the main source of food to over half of the world. Planting rice on rice accounts 34 percent of the total arable land in Sri Lanka and sustains over 20 million people in the island [1]. Although rice growing is of primary importance, it is still a problem, which struggles to achieve high efficiency in resources, maximum yield, and balanced land management. The classical methods of field management are based on the consistent application of inputs - the same amount of seed, fertilizer, and water on the whole field without taking into consideration the spatial differences in the quality of the soil and the health state of crops.

The modern precision agriculture acknowledges the fact that rice fields are not homogenous. There are huge spatial variations in the level of soil moisture, availability of soil nutrients, crop development and distribution of diseases in various parts of the same field. An attempt to use one management strategy in such a spatially heterogeneous environment will result in over fertilization in healthy areas, under treatment in stressed areas and wasteful use of seeds and agrochemicals. Precision agriculture research has repeatedly demonstrated that zone-based management; breaking up the field into spatially isolated management zones and using specialized inputs in each zone can enhance crop production, lower inputs cost and decrease environmental impact [2].

Although there are notable benefits of zone-based field management, it is hardly adopted by smallholder rice farmers. The major obstacle is the challenge in an accurate description of the spatial distribution of crop health and soil state at the field level. Manual scouting is costly, time-consuming and not feasible in a systematic manner of collecting data. Some of the benefits of remote sensing with either aerial drones or satellites may be realized but are generally not economical to small-scale farmers in developing nations, and the canopy density of flooded rice paddies may restrict the capability of aerial images to identify ground-based soil conditions [3].

The Smart Rice Field Intelligence System can overcome these constraints by applying an autonomous ground rover as the main sensing platform. The rover traverses right across the rice field and takes measurements of GPS quality indicators, soil measurements and plant level imagery. The measurements are processed by four composite parts, namely, GPS degradation monitoring (Component 1), AI-based crop health analysis (Component 2), spatial zone mapping and treatment planning (Component 3), and predictive soil intelligence (Component 4).

The study will be based on Component 3: Rice Field Zone Mapping and Autonomous Seeding and Treatment Planning. The AI analysis system provides the component with crop health intelligence, whereas the soil intelligence system provides it with predictions of the soil condition. This system uses these multi-source inputs to group the rice field into separate management zones, depending on agronomic conditions, using geospatial clustering algorithms. A suggested management intervention is then placed on each zone and can be one

of adjusted seeding density, specific fertilizer application, further irrigation or specific crop protection. The final product is a collection of spatially resolved zone maps and operation plans that are used to direct field interventions.

This study works well in eliminating the research gap that exists between the raw sensor data that is obtained by the rover and the decision support that is practical and ready to use by rice farmers in the field. The capacity to convert a multi-sensor data into well-defined spatial areas and clear management prescriptions is the key to the availability and practicality of precision agriculture to smallholder farmers. The system complies with the United Nations Sustainable Development Goals (SDGs): SDG 2 (Zero Hunger) as it will enhance the productivity of rice, SDG 12 (Responsible Consumption and Production) as it will decrease the wasteful usage of farmed inputs, and SDG 15 (Life on Land) as it will reduce the environmental impact of agricultural practices [4].

1.2 Literature Review

Precision agriculture and variable-rate field management have been extensively studied over the past two decades. The core principle -that spatially variable inputs should be applied according to site-specific conditions - is well established in both agronomic and engineering literature.

Mulla [5] provided a comprehensive review of precision agriculture technology, emphasizing that spatial data collection and variable-rate application form the foundation of modern crop management. The review highlighted that dividing agricultural fields into management zones based on measured variability in soil and crop properties is a practical and scalable approach to precision farming, especially for resource-constrained settings.

Zone-based field management has been demonstrated across multiple crop types. Moral et al. [6] applied K-means clustering to soil nutrient datasets collected across wheat fields, generating distinct management zones with statistically significant differences in soil pH, nitrogen, and phosphorus levels. Their results showed that variable-rate fertilizer application based on zone maps reduced fertilizer use by up to 22% while maintaining equivalent crop yields. This work demonstrates that unsupervised clustering methods are appropriate for agricultural zone delineation, which directly informs the algorithmic approach in this research.

For rice-specific precision agriculture, Balasubramanian et al. [7] demonstrated that site-specific crop management, including spatially adjusted seeding rates and nutrient applications, significantly improved yields in irrigated lowland rice systems in South and Southeast Asia. Their field studies showed that yield gaps of 20-30% can be closed through proper zone-based management, underscoring the value of the spatial planning functionality proposed in this research.

Geospatial data processing for agricultural applications has been substantially advanced by developments in open-source GIS tools. Congalton and Green [8] reviewed accuracy assessment methods for geospatial data, providing validated frameworks for evaluating the reliability of zone classification outputs. Their work on confusion matrices and kappa coefficients provides the methodological basis for validating the zone classification accuracy in this research.

Vegetation indices derived from imagery have become standard inputs for crop health zone mapping. Xue and Su [9] conducted a comprehensive review of vegetation indices applied to crop monitoring, demonstrating that indicators such as NDVI (Normalized Difference Vegetation Index) provide reliable proxies for crop biomass, chlorophyll content, and stress levels. These indices, derived from the crop health analysis component, serve as primary inputs for zone classification in this research.

Seeding rate optimization has been addressed in multiple studies on paddy cultivation. Bouman et al. [10] investigated optimal seeding densities in direct-seeded rice systems, concluding that recommended seeding rates should vary based on soil condition, expected germination rate, and prior crop performance. The rule-based seeding recommendation framework in this research draws on similar agronomic principles to derive density recommendations for each identified field zone.

Treatment planning and intervention scheduling in precision agriculture have been reviewed by Gebbers and Adamchuk [11], who emphasized that effective decision support systems must combine real-time field measurements with agronomic knowledge to generate actionable recommendations. Their framework for sensor-based variable rate application directly informs the treatment planning logic in this component.

Geographic Information Systems (GIS) have long been used for agricultural mapping. Papadopoulos et al. [12] demonstrated that GIS-based visualization tools significantly improve the usability of precision agriculture outputs for farm managers by presenting complex geospatial data in intuitive, interpretable formats. The visualization dashboard proposed in this research follows this design approach.

Despite these advances, a significant gap exists in systems designed for smallholder rice farming in developing regions. Most precision agriculture systems described in the literature are designed for large-scale mechanized farms with existing GIS infrastructure [13]. There is limited work on lightweight, accessible zone mapping systems that can operate on

sensor data collected by low-cost ground-based rovers without requiring expensive airborne imagery or commercially licensed GIS platforms.

1.3 Research Gap

A review of existing precision agriculture systems and rice field management tools reveals several important gaps that this research aims to address.

First, most existing zone mapping systems rely on aerial imagery from drones or satellites as their primary data source. While effective in large open fields, aerial imagery is less reliable in flooded paddy environments where reflectance from standing water can distort vegetation index calculations, and where ground-level soil conditions cannot be directly observed from above. Very few systems integrate ground-level soil sensor data with crop imagery for zone delineation, leaving a methodological gap that this research fills by using rover-collected multi-sensor data.

Second, the majority of agricultural decision support systems in the literature operate on historical or static field data rather than integrating real-time or session-based measurements. The proposed system is designed to receive fresh inputs from the rover's current field survey, enabling up-to-date zone classification that reflects current crop and soil conditions rather than relying on fixed historical baselines.

Third, existing systems rarely include an integrated seeding recommendation engine alongside a treatment planning system. Seeding density recommendations and fertilizer/treatment plans are typically produced by separate specialist tools that require significant technical expertise to operate. This research proposes a unified component that produces both seeding and treatment recommendations from the same zone classification framework.

Fourth, accessible visualization tools tailored for smallholder farmers in developing agricultural contexts are largely absent from the literature. Most GIS visualization outputs require trained GIS analysts to interpret. The visualization dashboard in this research is designed to present field zones and recommendations in a non-technical, farmer-friendly format.

These gaps justify the development of an integrated zone mapping and planning system that processes ground rover sensor data to generate practical rice field management recommendations.

Feature	[5]	[6]	[7]	[10]	[11]	[13]	Proposed System
Ground rover data input	x	x	x	x	x	x	✓
Spatial zone clustering	✓	✓	Limited	x	x	x	✓
Seeding recommendation	x	x	✓	✓	x	x	✓
Treatment planning	Limited	x	x	x	✓	✓	✓
Soil + crop data fusion	x	✓	x	x	x	x	✓
Mission planning map	x	x	x	x	x	x	✓
Farmer-friendly dashboard	x	x	x	x	x	x	✓
Open-source GIS tools	x	Limited	x	x	x	x	✓

Table 1-System Comparison

2. OBJECTIVE

2.1 Main Objectives

To design, develop, and evaluate a spatial analysis and planning system that converts crop health intelligence and soil condition measurements, collected by an autonomous ground rover,

into structured rice field management zones and actionable seeding and treatment recommendations that support precision rice cultivation.

The system will be implemented and evaluated during the research period, targeting a zone classification accuracy of at least 80% compared to expert agronomic evaluation, and demonstrating consistent and agronomically valid seeding and treatment recommendations during pilot testing.

2.2 Specific Objectives

1. Field Zone Segmentation and Classification

Develop a spatial zone segmentation algorithm that divides the rice field into distinct management zones using multi-source inputs including crop health indicators (from Component 2) and soil condition predictions (from Component 4). The segmentation algorithm will apply K-means clustering and DBSCAN spatial density methods to identify zones with statistically distinct agronomic profiles. Each zone will be classified according to a defined agronomic condition framework covering healthy crop zones, moderate stress zones, severe stress zones, and soil-deficient zones.

2. Seeding Recommendation Generation

Design and implement a rule-based seeding recommendation engine that generates optimal seeding density recommendations for each field zone based on soil quality classification, prior crop performance data, and zone-specific agronomic conditions. Recommendations will distinguish between areas requiring standard seeding density, areas requiring gap-filling reseeding due to poor germination, and areas where reduced seeding is appropriate due to high competition or waterlogging risk.

3. Treatment Planning and Intervention Scheduling

Develop a treatment planning module that assigns appropriate agricultural interventions to each classified zone. Recommended interventions include targeted fertilizer application (nitrogen, phosphorus, and potassium adjustments), supplemental irrigation recommendations for moisture-stressed zones, crop protection measures for disease-affected zones, and monitoring-only designations for healthy zones. Intervention priorities will be ranked by severity and operational urgency.

4. Mission Planning Map Generation

Generate georeferenced spatial maps that present field zone boundaries, zone classifications, recommended seeding areas, and prioritized treatment locations in a format suitable for guiding agricultural field operations. Maps will be produced in standard geospatial formats compatible with GIS tools and portable display devices used in field settings.

5. Visualization Dashboard Development

Design and implement a visualization dashboard that presents field zone maps, soil condition overlays, crop health distributions, seeding recommendations, and treatment priorities in a clear and non-technical format. The dashboard is intended for use by farmers, agricultural extension officers, and field supervisors who may not have GIS expertise.

3. METHODOLOGY

Research Approach

This research follows a Design Science Research (DSR) methodology, which is appropriate for projects focused on designing, building, and evaluating a novel technological artifact. The primary goal is to develop a functional spatial planning system and evaluate its effectiveness in producing accurate zone classifications and agronomically valid recommendations. An experimental evaluation component compares system-generated zone maps and recommendations against expert agronomic assessment and standard field management guidelines.

The development process follows an iterative cycle: requirement gathering from agronomic literature and expert consultation, algorithm design, software implementation, testing with simulated and real datasets, and validation against agronomic benchmarks.

Technical Methods and Algorithms

Data Input and Preprocessing

The component receives two primary data streams from the integrated rover system. The first stream is crop health data from Component 2, including NDVI values, disease detection flags, growth stage classifications, and stress probability scores for spatially referenced field locations. The second stream is soil condition predictions from Component 4, including soil moisture levels, soil pH, nitrogen and phosphorus indicator scores, and predicted soil quality classifications per location.

Raw data is preprocessed to handle missing values, normalize sensor readings to a common scale, and align spatial coordinates. Data cleaning removes outlier measurements caused by sensor noise or GPS positioning errors. The output of preprocessing is a structured geospatial dataset where each sampled field location is represented by a feature vector containing crop health and soil condition descriptors.

Field Zone Segmentation Algorithm

Zone segmentation is performed using two complementary clustering approaches. K-means clustering is applied as the primary algorithm, with the number of clusters determined through silhouette score analysis and domain knowledge (typically three to five zones for a standard rice field). K-means partitions field locations into zones that minimize within-cluster variance across crop and soil feature dimensions.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is applied as a secondary method to identify spatially contiguous clusters and handle irregular field geometries. DBSCAN does not require a predefined number of clusters, making it useful for discovering unexpected spatial patterns in heterogeneous fields. The outputs of both algorithms are compared, and agronomically coherent zone boundaries are selected through expert review.

Zone Classification Framework

Each identified cluster is assigned an agronomic zone classification using a rule-based decision framework. Classification rules are defined based on threshold values for key indicators, informed by agronomic literature and expert consultation. The classification categories are: Zone A (Healthy — high NDVI, optimal soil conditions), Zone B (Moderate Stress — reduced NDVI or mild soil deficiency), Zone C (Severe Stress — low NDVI, significant disease or soil problem), and Zone D (Soil-Deficient — below-threshold nutrient levels with limited crop response).

Rule-Based Seeding Recommendation Engine

Seeding recommendations are generated by mapping zone classifications and soil quality scores to predefined seeding density categories. The decision logic applies thresholds derived from standard rice cultivation guidelines. For each zone, the engine outputs a recommended seeding density (in kg/ha), a planting method recommendation (transplanting vs direct seeding), and a gap-filling priority flag where existing crop stand density is below acceptable thresholds.

Treatment Planning Logic

Treatment plans are generated through a zone-specific decision tree. Each zone classification triggers a set of recommended interventions. Zone A zones receive a monitoring-only designation. Zone B zones receive moderate fertilizer adjustment recommendations. Zone C zones receive high-priority treatment flags including targeted fungicide or pesticide application if disease is detected, plus fertilizer correction. Zone D zones receive soil amendment recommendations and possible drainage or irrigation interventions. All recommendations are prioritized by expected impact on crop recovery and operational urgency

Tools and Platforms with Justification

The system is implemented in Python due to its strong ecosystem of geospatial and data analysis libraries. The core geospatial processing library is GeoPandas, which provides efficient handling of spatial data in GeoDataFrame format and supports spatial join, overlay,

and geometric operations required for zone boundary delineation. Scikit-learn is used for implementing K-means and DBSCAN clustering algorithms, providing well-tested and computationally efficient implementations.

Matplotlib and Folium are used for visualization output. Matplotlib generates static zone maps suitable for printing and field use, while Folium generates interactive HTML-based maps for use in the dashboard interface. NumPy and Pandas handle numerical data processing and tabular data management throughout the pipeline.

QGIS is used as the reference GIS environment for validating zone boundaries and georeferenced outputs. The system is designed to export outputs in GeoJSON format, which is compatible with QGIS and other standard GIS tools. A Flask-based lightweight web interface serves the visualization dashboard, making it accessible through standard web browsers on any device.

Git and GitHub are used for version control and collaborative development. Visual Studio Code serves as the primary development environment.

Data Collection and Ethical Considerations

Data for this research will be collected from three sources. First, publicly available agricultural datasets from research institutions such as the International Rice Research Institute (IRRI) and FAO will be used to develop and test the clustering algorithms. These datasets include soil nutrient measurements, rice yield records, and field management data from multiple growing seasons across different rice cultivation regions.

Second, simulated rover data will be generated based on published rice field sensor studies to test the full data pipeline from input ingestion through zone classification to recommendation output. Simulation parameters will be derived from the literature to represent realistic rice field conditions.

Third, agronomic expert consultation will be conducted with agricultural scientists and extension officers to validate zone classification rules, seeding density recommendations, and treatment planning thresholds. Expert feedback will be gathered through structured interviews.

All data collection activities will follow ethical research guidelines. Expert participants will provide informed consent before interviews. No personal data about farmers or private landowners will be collected.

Validation Strategy and Performance Metrics

The system will be validated against expert-defined ground truth zone maps for a set of test field scenarios. Validation will assess both zone classification accuracy and the agronomic quality of seeding and treatment recommendations.

Zone classification accuracy will be measured using overall accuracy and Cohen's kappa coefficient, computed by comparing system-generated zone classifications with expert-defined reference classifications for the same spatial locations. A target accuracy of 80% or higher is set for the final system.

Seeding recommendation consistency will be evaluated by comparing system outputs against agronomic standard tables for rice cultivation, measuring the percentage of recommendations falling within acceptable agronomic ranges. Treatment planning reliability will be assessed by agronomic experts rating the appropriateness of recommended interventions for each test zone.

Computational performance will be evaluated by measuring zone map generation time for various field sizes, ensuring the system can produce outputs within a practical timeframe for operational use. A target of under five minutes for a standard field size of two hectares is set.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

The following diagrams illustrate the overall system architecture and component-level design for the Rice Field Zone Mapping and Autonomous Seeding and Treatment Planning system.

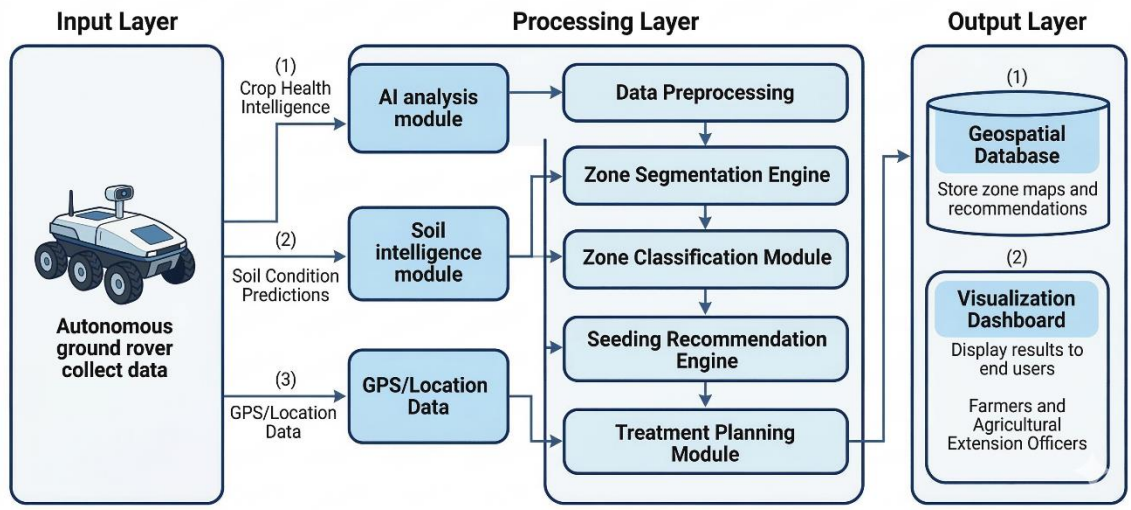


Figure 1-system diagram

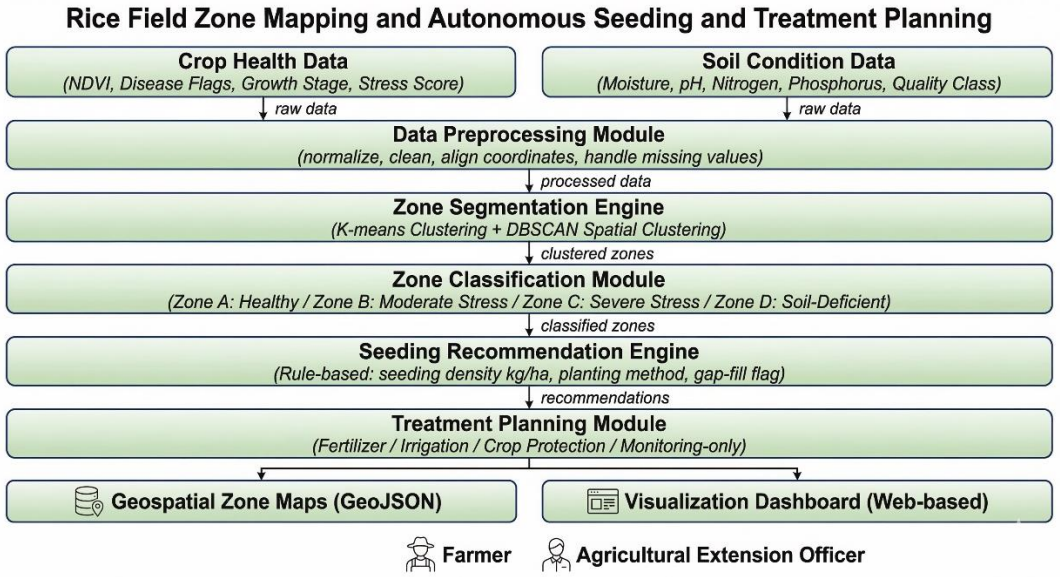


Figure 2-Component Diagram

The system receives multi-source inputs from the rover data pipeline and processes them through five core modules: Data Preprocessing, Zone Segmentation Engine, Zone Classification Module, Seeding Recommendation Engine, and Treatment Planning Module. Outputs are stored in a geospatial database layer and presented through the Visualization Dashboard.

5.USER REQUIREMENTS

Stakeholder identification

The proposed zone mapping and planning system involves several key stakeholder groups.

Rice Farmers (Primary Users): Farmers operating rice cultivation fields are the primary beneficiaries of the system. They interact with the visualization dashboard to view zone maps and receive seeding and treatment recommendations in a practical, actionable format.

Agricultural Extension Officers (Secondary Users): Government extension officers and agronomists use the system outputs to advise farmers on field interventions. They require access to more detailed zone reports and recommendation rationales.

Research Team (Tertiary Users): The research team uses the system for testing, validation, and algorithm refinement. They require access to raw processing outputs, clustering diagnostics, and performance metrics.

Functional Requirements

1. The system shall ingest crop health data and soil condition data from the rover data pipeline in standard CSV or GeoJSON format.
2. The system shall apply clustering algorithms to divide the field into a minimum of three and maximum of six management zones.
3. The system shall classify each zone according to the defined agronomic condition framework.
4. The system shall generate seeding density recommendations for each zone based on zone classification and soil quality scores.
5. The system shall generate a prioritized treatment plan for each zone including specific intervention types.
6. The system shall produce georeferenced zone maps in GeoJSON format compatible with standard GIS tools.
7. The system shall display zone maps, soil overlays, and recommendations through a web-based visualization dashboard.
8. The system shall allow users to export zone maps and recommendations as PDF reports.

Non-Functional Requirements

Accuracy: Zone classification accuracy shall exceed 80% against expert-evaluated ground truth.

Performance: Zone map generation shall complete within five minutes for a two-hectare field on standard hardware.

Usability: The dashboard interface shall be operable by non-technical users with minimal training.

Reliability: The system shall process all input records without data loss or silent failure.

Portability: The system shall run on standard desktop or laptop hardware without specialized GIS infrastructure.

Scalability: The architecture shall support field sizes of up to 10 hectares without significant performance degradation.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market is smallholder rice farming households across South and Southeast Asia, particularly in Sri Lanka, India, Vietnam, and the Philippines, where rice is the dominant staple crop and precision agriculture adoption remains limited. Secondary markets include government agricultural extension services, agribusiness companies providing advisory services to smallholder farmers, and agricultural research institutions developing precision farming tools.

Value Proposition

Value Element	Description
Reduced Input Waste	Variable-rate seeding and treatment recommendations reduce seed and fertilizer waste by targeting inputs to zones that need them.
Improved Yield	Zone-based management addresses spatial variability, closing yield gaps that uniform management leaves unaddressed.
Practical Field Maps	Georeferenced zone maps guide field operations directly, reducing guesswork and manual scouting effort.
Farmer-Friendly Dashboard	Non-technical visualization of zone status and recommendations is accessible without GIS expertise.
Integrated with Rover Data	Directly consumes rover sensor outputs without requiring expensive aerial imagery or commercial GIS platforms.
Extensible Architecture	Modular design supports future integration with additional sensors, satellite imagery, or IoT field devices.
Low Infrastructure Cost	Open-source tools eliminate software licensing costs, making adoption viable for smallholder contexts.

Table 2-Value Proposition

Revenue Model

A hybrid revenue model is proposed for commercialization.

- Software-as-a-Service (SaaS) Subscription: Monthly or annual subscription for farmers and cooperatives, providing access to the zone mapping and planning platform.
- Institutional Licensing: Annual licenses for government agricultural agencies and NGOs deploying the system across multiple farms.
- B2B Data Services: Aggregated field analytics reports for agribusiness companies and agricultural input suppliers who benefit from understanding regional crop condition patterns.

Consulting and Integration Services: Technical support and integration services for organizations deploying the system alongside existing farm management platforms.

Cost Estimation

Cost Category	Estimated Cost (LKR)	Details
Software Development	Rs. 30,000 – 45,000	Python geospatial pipeline, clustering modules, dashboard development.
Expert Consultation	Rs. 10,000 – 15,000	Agronomic expert interviews for algorithm validation and recommendation rule design.
Data Collection & Licensing	Rs. 5,000 – 8,000	Acquisition of agricultural datasets for algorithm training and testing.
Testing and Validation	Rs. 8,000 – 12,000	Pilot testing, usability evaluation, performance measurement.
TOTAL INITIAL COSTS	Rs. 53,000 – 80,000	

Table 3-Cost Estimation

Pricing Strategy

A penetration pricing strategy is recommended to maximize adoption among resource-constrained smallholder farmers. Initial pricing will be set below market rates for comparable precision agriculture services, with pricing tiers for individual farmers, cooperative groups, and institutional users.

Competitive Advantage

The proposed system offers unique competitive advantages over existing precision agriculture platforms. It is specifically designed for ground-rover-based data inputs rather than aerial imagery, making it more appropriate for flooded paddy environments. It integrates seeding and treatment planning in a single unified pipeline, which is absent from most existing tools. It operates entirely on open-source software, eliminating licensing barriers for adoption in developing regions.

Intellectual Property Considerations

The intellectual property of the proposed system includes the software architecture, the zone classification rule framework, the seeding recommendation algorithm, and the visualization dashboard design. Copyright protection will be applied to all developed software code and

documentation. The rule-based recommendation logic and clustering configuration framework may be documented as a technical methodology contribution through academic publication.

7.BUDGET AND JUSTIFICATION

Cost Category	Estimated Cost (LKR)	Justification
Travel Expenses	9,000	Visits to SLIIT campus for supervisor and co-supervisor meetings. Visits to agricultural research stations and field sites for expert consultation.
Data Collection	8,000	Expert consultation appointments with agricultural scientists and agronomists. Access fees for agricultural datasets.
Printing & Documentation	1,500	Printing of proposals, reports, consent forms, and survey instruments.
Computing Resources	2,000	Cloud computing credits for large-scale algorithm testing and geospatial processing.
Total	20,500	

Table 4-Budget and Justification

8.WORK BREAKDOWN STRUCTURE (WBS)

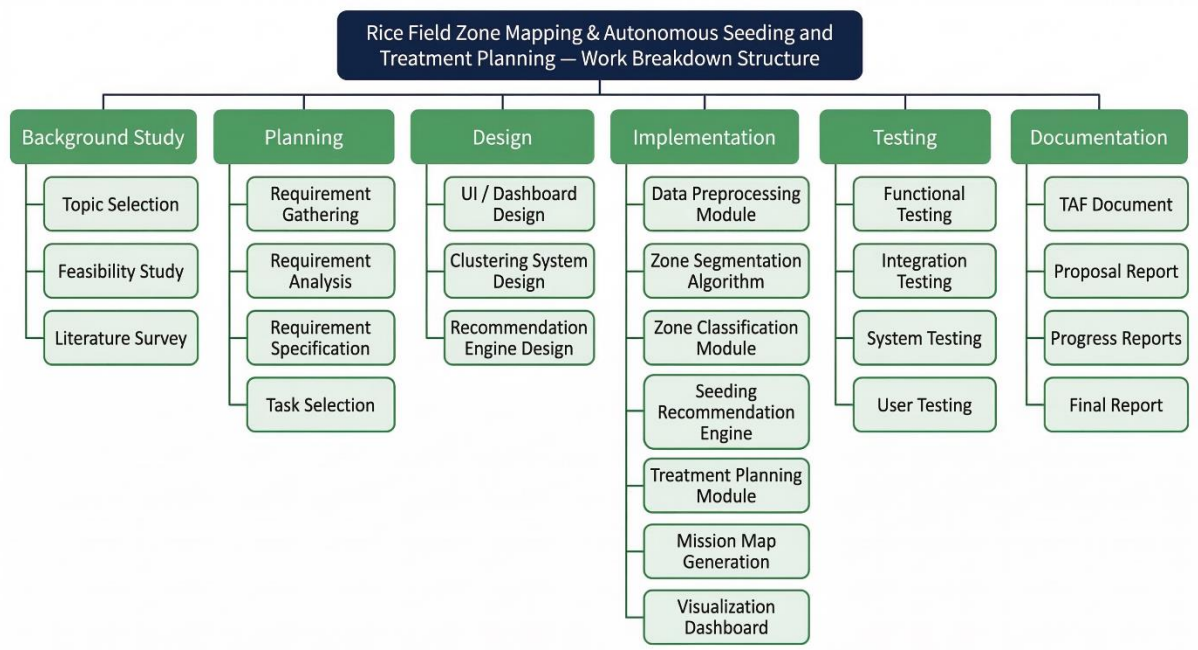


Figure 3-Budget and Justification

9. GANTT CHART

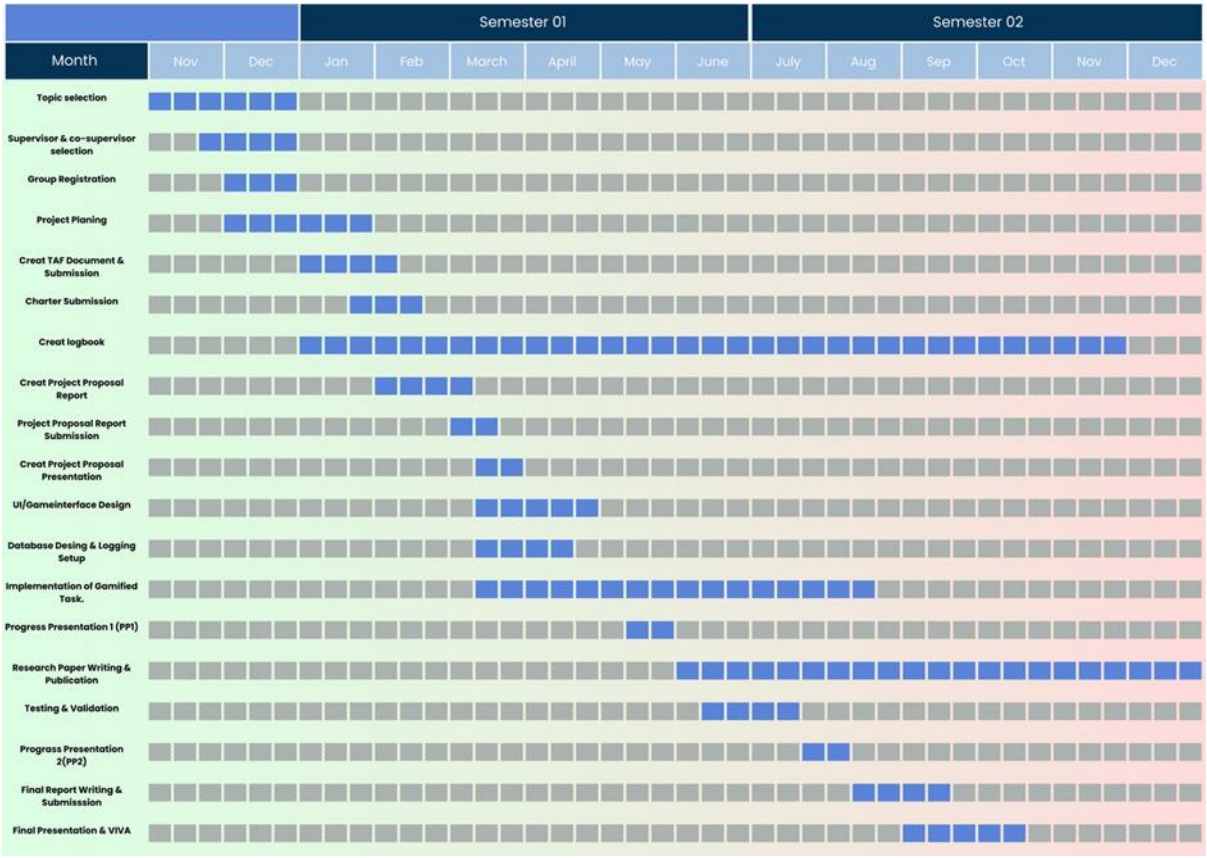


Figure 4-Gantt Chart

10 .REFERENCES LIST

11.APPENDICES

Survey instruments

Rice Field Zone Mapping and Autonomous Seeding and Treatment Planning - Expert Consultation Questionnaire

Introduction

This questionnaire is part of a fourth-year research project focused on developing a spatial zone mapping and precision treatment planning system for rice fields. The system processes crop health and soil condition data collected by an autonomous ground rover to generate field management zones and seeding or treatment recommendations.

We are seeking expert insights from agronomists, agricultural scientists, and rice cultivation specialists to inform the design of the zone classification rules and recommendation thresholds. All responses will be used solely for academic research purposes and kept confidential.

Questions

1. In your experience, what are the most important indicators used to classify rice field zones by agronomic condition (e.g., soil pH range, NDVI thresholds, moisture levels)?
2. What are the typical seeding density recommendations (kg/ha) for different soil quality levels in Sri Lankan rice varieties (e.g., BG 300, Samba Masuri)?
3. When assessing a rice field zone as requiring priority treatment, what combination of factors most commonly triggers an immediate intervention decision?
4. How do you differentiate between a zone that needs fertilizer correction and one that requires drainage improvement, based on observable field indicators?
5. Are there established local agronomic standards or guidelines (from DOA or IRRI) that should govern treatment type and rate recommendations for low-resource farming contexts?
6. What geospatial resolution is practically sufficient for field zone delineation in a two-hectare paddy field — by plot, by row, or by GPS grid cell?
7. What types of visualization would be most useful for farmers or extension officers using a zone map in the field (e.g., color-coded maps, tabular summaries, annotated aerial views)?
8. Are there any zone classification approaches or seeding recommendation systems you are aware of that have been validated in Sri Lankan rice cultivation contexts?

9. From your professional experience, what is the most common mistake made in uniform-input rice farming that spatial zone management could correct?
10. What one feature would you most want to see in a decision support system designed to help smallholder rice farmers manage their fields more precisely?From your professional experience, what is the most common failure mode in UAV-based precision spraying operations that should be prioritized in simulation testing?

Risk analysis

Risk Factor	Impact	Probability	Mitigation Strategy
Inaccurate zone classification (clustering algorithm produces agronomically invalid zones)	High	Medium	Validate clustering outputs against expert-defined reference zones. Use silhouette scoring and expert review to tune cluster parameters. Apply both K-means and DBSCAN and select the best output per field scenario.
Insufficient or low-quality input data from rover components	High	Low	Design robust preprocessing with missing value handling and outlier removal. Test with synthetic datasets to ensure the pipeline degrades gracefully with sparse data.
Seeding recommendations outside agronomic acceptable ranges	High	Low	Define recommendation thresholds collaboratively with agronomic experts. Implement hard constraint checks that flag out-of-range outputs before presentation.
Geospatial coordinate misalignment between crop and soil datasets	Medium	Medium	Apply coordinate alignment and spatial join validation during preprocessing. Implement tolerance-based spatial matching to handle minor GPS positional errors.
Dashboard usability issues for non-technical users	Medium	Medium	Conduct usability testing with representative users including farmers and extension officers. Apply iterative UI refinement based on feedback.
Computational performance issues with large field datasets	Medium	Low	Profile algorithm performance with increasing dataset sizes. Implement data sampling strategies for very large fields to maintain acceptable processing time.

Table 5-Risk Analysis

SMART RICE FIELD INTELLIGENCE SYSTEM USING AN AUTONOMOUS GROUND ROVER

Component 04: Predictive Soil Intelligence and Rice Field Decision Support

R26-IT-75

Project Proposal Report

S.N.R Jayawardhana – IT22192950

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

Classification: Public-SLIIT

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
S.N.R Jayawardhana	IT22192950	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Samantha Rajapaksha		

Classification: Public-SLIIT

ABSTRACT

Rice cultivation is the backbone of food security across South and Southeast Asia, yet small-scale rice farmers continue to face critical challenges in maintaining optimal soil conditions, managing water resources, and making timely crop management decisions. In Sri Lanka alone, rice accounts for over 34% of total agricultural land, yet average yields remain significantly below potential due to poor soil health monitoring, inappropriate fertilizer application, and the absence of data-driven decision support tools accessible to ordinary farmers.

This research proposes an intelligent, sensor-driven Predictive Soil Intelligence and Rice Field Decision Support System as Component 04 of the broader Smart Rice Field Intelligence System using an Autonomous Ground Rover. The system integrates IoT-based soil sensor arrays (measuring pH, moisture, nitrogen, phosphorus, potassium, and temperature) with satellite remote sensing data, weather API feeds, and farmer-reported observations to construct a comprehensive, real-time understanding of rice field conditions.

A machine learning prediction engine, trained on historical soil and yield datasets, continuously forecasts soil nutrient depletion, waterlogging risk, and optimal harvest windows. A rule-based crop advisory engine translates these predictions into actionable recommendations for fertilizer scheduling, irrigation timing, and pest/disease alerts. All insights are delivered through a farmer-friendly mobile application and a web-based agronomist dashboard, with SMS and push notification alerts for time-critical decisions.

The platform is developed using React Native for the mobile interface, Node.js with Express for backend services, MongoDB and Firebase for data management, and Python-based ML models (scikit-learn, TensorFlow Lite) for predictive analytics. Expected outcomes include real-time soil health classification with over 85% accuracy, measurable improvements in fertilizer efficiency, and farmer-accessible decision support that reduces crop loss and improves yield.

Keywords: Digital Twin, Precision Agriculture, Predictive Analytics, IoT Sensors, Decision Support System.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of my fourth-year research project.

First and foremost, I would like to thank our Research Project Supervisor Prof. Samantha Rajapaksha for continuous guidance, encouragement, and valuable support throughout the entire research process. Their advice, expertise, and constructive feedback played a significant role in the successful development of this proposal.

I would also like to extend my sincere appreciation to our Co-Supervisor Ms. Chathurya Prabhavi Kumarapperumafor valuable insights, suggestions, and guidance provided during the planning stages of this research.

Classification: Public-SLIIT

I would like to extend special thanks to the agricultural experts, soil scientists, and experienced rice farmers in the Kurunegala and Polonnaruwa districts of Sri Lanka who generously shared their practical knowledge and field experience. Their insights were invaluable in understanding the real-world challenges of soil management and rice cultivation.

My sincere appreciation also goes to the Department of Agriculture Sri Lanka for providing access to historical soil and yield data that guided the design of the machine learning components of this system.

I would also like to thank my group members for their teamwork, cooperation, and dedication throughout this project. Their collaboration and support were essential in completing the overall system architecture successfully.

I would also like to thank the academic and technical staff at the Department of Information Technology, Sri Lanka Institute of Information Technology, for providing access to computational resources, simulation environments, and research support.

Finally, I would like to express my heartfelt gratitude to my family and friends for their encouragement, patience, and continuous support during the entire research journey.

TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 3
TABLE OF CONTENT 4
LIST OF TABLES 5
LIST OF FIGURES 6
LIST OF ABBREVIATIONS 7
LIST OF APPENDICES 7
1.INTRODUCTION 8
1.1 Background and Context 8
1.2 Literature Review 9
1.3 Research Gap 10
2. OBJECTIVE 11
2.2 Specific Objectives 11
3. METHODOLOGY 13
Research Approach 13
Technical Methods and Algorithms 13
Tools and Platforms with Justification ..... **Error! Bookmark not defined.**
Data Collection and Ethical Considerations 13
Validation Strategy and Performance Metrics 15
4.HIGH-LEVEL SYSTEM ARCHITECTURE 16
5.USER REQUIREMENTS 17
Stakeholder identification 17

Classification: Public-SLIIT

Functional Requirements 17
Non-Functional Requirements 18
6. COMMERCIALIZATION PLAN 19
Target Market and Customer Persona 19
Value Proposition 19
Revenue Model 21
Cost Estimation 21
Pricing Strategy 22
Competitive Advantage 22
Intellectual Property Considerations 22
7.BUDGET AND JUSTIFICATION 22
8.WORK BREAKDOWN STRUCTURE (WBS) 23
9. GANTT CHART 23
10 .REFERENCES LIST 24
11.APPENDICES 24
Survey instruments 25
Plagiarism Report ..... **Error! Bookmark not defined.**
Risk analysis ..... **Error! Bookmark not defined.**

LIST OF TABLES

Table 01 - System Comparison …….. 14

Table 02 - value proposition……... 24

Table 03 - Cost Estimation…… 25

Table 04 - Budget and Justification……... 26

Table 05 - Gantt Chart……..……...23

Table 06-Risk Analysis……26

Classification: Public-SLIIT

LIST OF FIGURES

Figure 01 - System diagram ……..... 21

Figure 02 - Component diagram …….……..... 21

Figure 03 - Work Breakdown Chart …….. 29

Classification: Public-SLIIT

LIST OF ABBREVIATIONS

Abbreviation	Word / Description
IoT	Internet of Things
ML	Machine Learning
AI	Artificial Intelligence
NPK	Nitrogen, Phosphorus, Potassium
REST	Representational State Transfer
API	Application Programming Interface
WPM	Words Per Minute (not applicable here – kept for template parity)
DSR	Design Science Research
UI	User Interface
SDK	Software Development Kit
SMS	Short Message Service
GIS	Geographic Information System
SDGs	Sustainable Development Goals
IP	Intellectual Property

LIST OF APPENDICES

1.Survey instruments ..... **Error! Bookmark not defined.**
2.Plagiarism Report ..... **Error! Bookmark not defined.**
3.Risk analysis ..... **Error! Bookmark not defined.**

Classification: Public-SLIIT

1.INTRODUCTION

1.1 Background and Context

Rice is the single most important food crop in Sri Lanka and across much of Asia, providing the primary caloric intake for billions of people. In Sri Lanka, rice cultivation occupies approximately 34% of total agricultural land and directly supports the livelihoods of over 1.8 million farming families. Despite its central role in national food security, average paddy yields in Sri Lanka remain at around 4.2 metric tons per hectare — significantly below the achievable potential of 6–8 metric tons per hectare under optimally managed conditions [1]. This yield gap is not simply a result of climate variability or inadequate water supply; rather, it is fundamentally driven by poor soil health management, inefficient fertilizer use, and the near-total absence of real-time, data-driven decision support tools accessible to smallholder farmers.

Soil is the foundational resource of agricultural production. The physical, chemical, and biological properties of soil — including pH, moisture content, organic matter, and macronutrient concentrations (NPK: Nitrogen, Phosphorus, Potassium) — directly determine the capacity of a field to support healthy rice growth and maximize yields. In practice, however, the majority of Sri Lankan smallholder. farmers rely almost entirely on conventional wisdom, seasonal experience, or generalized government advisories when making critical soil management decisions. Laboratory soil testing, where it exists, is conducted infrequently and its results are rarely translated into timely, field-specific recommendations [2].

The emergence of precision agriculture technologies — including IoT-based sensor networks, satellite remote sensing, unmanned aerial vehicles (drones), and machine learning — offers an unprecedented opportunity to transform rice farming from a practice based on generalised knowledge into one driven by real-time, field-specific intelligence. These technologies enable continuous monitoring of soil conditions across individual fields, predictive modelling of nutrient depletion and waterlogging risk, and automated, personalised recommendations delivered directly to the farmer's mobile device [3].

Despite the availability of these technologies globally, their adoption in smallholder rice farming in South Asia remains extremely limited. Existing precision agriculture systems are predominantly designed for large-scale commercial farms in high-income countries. They require expensive hardware installations, stable internet connectivity, and high levels of technical literacy — conditions that are rarely met in rural Sri Lanka. Additionally, most available systems provide raw data outputs without context-sensitive interpretation, making them inaccessible to farmers without agronomic expertise [4].

The proposed research, as Component 04 of the broader Smart Rice Field Intelligence System Using an Autonomous Ground Rover, directly addresses this gap by developing a Predictive Soil Intelligence and Rice Field Decision Support System. The system integrates low-cost IoT soil sensor arrays mounted on an autonomous ground rover, satellite imagery, weather data, and farmer-reported observations into a unified intelligence platform. A machine learning engine continuously analyses this multi-source data to predict soil nutrient status, waterlogging risk, and crop readiness, while a rule-based advisory engine translates these predictions into plain-language, actionable recommendations delivered through a mobile application.

Classification: Public-SLIIT

This solution directly supports the United Nations Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger) by improving agricultural productivity, SDG 9 (Industry, Innovation and Infrastructure) by introducing smart farming technology, and SDG 15 (Life on Land) by promoting sustainable soil management practices [5].

1.2 Literature Review

IoT-Based Soil Monitoring Systems

Research by Gubbi et al. [1] demonstrates that IoT sensor networks can enable real-time, continuous monitoring of soil conditions in agricultural settings, providing data accuracy superior to periodic laboratory testing. Studies integrating multiple soil sensors (pH, moisture, NPK, temperature) into unified monitoring platforms have shown significant improvements in fertilizer use efficiency and water management [6].

Machine Learning for Agricultural Prediction

Pantazi et al. [7] demonstrated the effectiveness of supervised machine learning algorithms — including Random Forests, Support Vector Machines (SVM), and Neural Networks — in predicting crop yield from multi-sensor soil data. Their models achieved prediction accuracy of up to 87% when trained on field-specific historical datasets. More recently, deep learning approaches using LSTM networks have shown promise for time-series prediction of soil moisture and nutrient dynamics [8].

Satellite Remote Sensing in Precision Agriculture

NDVI (Normalized Difference Vegetation Index) derived from satellite imagery has been widely validated as an indicator of crop health and biomass [9]. Integration of NDVI data with ground-level sensor readings has been shown to improve the accuracy of yield prediction models and enable early detection of stress conditions such as drought, waterlogging, and nutrient deficiency.

Decision Support Systems for Smallholder Farmers

Wolfert et al. [10] conducted a systematic review of digital agriculture platforms and found that decision support systems (DSS) combining multiple data sources with simplified user interfaces significantly improve smallholder farmer adoption and outcome quality. Mobile-first DSS platforms, designed for low-literacy users with visual dashboards and voice notifications, have demonstrated the highest adoption rates in South and Southeast Asian farming communities.

Crop Advisory and Recommendation Systems

Rule-based expert systems combined with machine learning have been successfully deployed in crop advisory applications. Rathore et al. [11] developed a rule-based fertilizer recommendation system for rice that achieved 82% agreement with expert agronomist recommendations, demonstrating the effectiveness of combining domain knowledge encoding with sensor-derived data.

Autonomous Rovers in Agriculture

The integration of autonomous ground vehicles with sensor payloads for in-field data collection has been explored in recent literature [12]. Rover-mounted soil sensors eliminate the need for fixed

Classification: Public-SLIIT

sensor grid installation, reduce per-field infrastructure costs, and enable dynamic, spatially-resolved soil mapping across heterogeneous field environments.

Limitations of Existing Systems

Despite significant advances, most existing precision agriculture systems share common limitations: high infrastructure cost, dependence on stable internet connectivity, insufficient localization for tropical rice varieties and soil types, and interfaces designed for technically literate users rather than smallholder farmers [13]. Furthermore, few systems integrate assessment, prediction, advisory, and progress monitoring within a single unified platform accessible via mobile devices.

1.3 Research Gap

Despite the growing body of research in precision agriculture and soil intelligence, several critical gaps remain, particularly in the context of smallholder rice farming in tropical developing countries such as Sri Lanka:

1. Lack of Integrated Multi-Source Platforms

Existing systems tend to operate on single data streams — either sensor data, satellite imagery, or weather data — but rarely integrate all three with farmer-reported observations into a unified, real-time intelligence engine.

2. Absence of Localized ML Models for Tropical Rice

Most published ML models for soil and crop prediction are trained on temperate-climate, large-scale farm datasets. Localised models trained on Sri Lankan paddy soil data, tropical weather patterns, and local rice varieties are largely absent from the literature.

3. Poor Accessibility for Low-Literacy Farmers

Technical outputs such as raw sensor readings, NDVI maps, or statistical predictions are not interpretable by most smallholder farmers. Existing platforms lack farmer-friendly interfaces that translate complex data into simple, actionable guidance in local languages.

4. No Closed-Loop Adaptive Advisory System

Current DSS platforms provide static or periodic recommendations rather than a closed-loop system that continuously monitors field response to interventions and dynamically adjusts future recommendations based on observed outcomes.

5. Limited Integration with Autonomous Rover Data

The emerging field of rover-based field sensing lacks integration with cloud-based ML prediction platforms and farmer-facing advisory systems, creating a gap between data collection and actionable intelligence.

Classification: Public-SLIIT

Feature	System A [7]	System B [10]	System C [11]	System D [12]	Proposed System
Real-time IoT Sensor Integration	✓	✗	✓	✓	✓
Satellite / NDVI Data Fusion	✗	✓	✗	✗	✓
ML-Based Yield Prediction	✓	✓	✗	✗	✓
Rule-Based Crop Advisory Engine	✗	✗	✓	✗	✓
Farmer Mobile App Interface	✗	✗	✗	✗	✓
SMS/Push Notification Alerts	✗	✗	✗	✗	✓
Closed-Loop Adaptive Advisory	✗	✗	✗	✗	✓
Rover-Mounted Sensor Support	✗	✗	✗	✓	✓
Agronomist Dashboard	✗	✓	✗	✗	✓
Localised for Tropical Rice	✗	✗	✗	✗	✓

Table 01 - System Comparison

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a Predictive Soil Intelligence and Rice Field Decision Support System that integrates real-time IoT sensor data, satellite imagery, and weather information with machine learning prediction models and rule-based advisory engines to deliver accurate, actionable, and farmer-accessible soil health assessments and crop management recommendations for smallholder rice farmers in Sri Lanka.

2.2 Specific Objectives

1. Real-Time Soil Health Monitoring and Classification

Develop and deploy an IoT sensor integration layer that continuously collects soil pH, moisture, NPK concentration, and temperature data from rover-mounted sensors. Implement a weighted scoring algorithm that classifies real-time field soil health into three categories — Optimal, Moderate Stress, or Critical — based on predefined agronomic thresholds validated by domain experts. Achieve classification accuracy greater than 85% compared to laboratory reference measurements.

Classification: Public-SLIIT

2. Machine Learning-Based Soil and Yield Prediction

Design and train predictive models using historical soil data, weather records, and satellite NDVI time-series to forecast: (a) soil nutrient depletion over the next 7–14 days, (b) waterlogging and drought stress risk, and (c) optimal harvest window prediction. Models will be implemented using scikit-learn (Random Forest, Gradient Boosting) and TensorFlow Lite for on-device inference capability, targeting prediction accuracy of ≥82% on held-out validation data.

3. Rule-Based Crop Advisory Engine

Develop a structured, expert-knowledge-encoded rule engine that translates ML prediction outputs and real-time sensor readings into specific, ranked, plain-language recommendations for fertilizer application type and quantity, irrigation scheduling, pest and disease early warning, and post-harvest preparation. Rules will be validated against the Sri Lanka Department of Agriculture rice cultivation guidelines and reviewed by at least three certified agronomists.

4. Farmer-Accessible Mobile Application and Alert System

Design and implement a React Native mobile application with a visual, low-text interface presenting soil health status, crop advisory messages, field maps, and historical trends in Sinhala and English. Integrate push notification and SMS alert systems for critical events such as waterlogging risk, urgent fertilizer needs, or optimal irrigation windows. Achieve a System Usability Scale (SUS) score above 75 in farmer usability trials.

5. Agronomist Web Dashboard and Reporting

Build a web-based agronomist dashboard that displays field-level soil intelligence maps, longitudinal soil health trends, prediction confidence metrics, and exportable PDF/CSV reports. Enable agronomists to configure advisory rule thresholds and monitor multiple fields simultaneously.

Classification: Public-SLIIT

3. METHODOLOGY

Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate given that the primary goal is to design, build, and evaluate a technological artefact — the Predictive Soil Intelligence System — that solves a clearly identified practical problem. The research follows an iterative build-evaluate-refine cycle, beginning with problem identification and requirements gathering, progressing through system design and development, and concluding with empirical evaluation against performance benchmarks.

he studies also incorporates an experimental evaluation component in which the system's soil classification outputs will be compared against laboratory reference measurements, and advisory recommendations will be validated against expert agronomist assessments. A pilot field trial with a small group of rice farmers will evaluate usability and real-world applicability.

Technical Methods and Algorithms

Multi-Sensor Data Acquisition

IoT sensors mounted on the autonomous ground rover collect soil data at configurable spatial intervals across the rice field. Sensors measure soil moisture (capacitive sensor), soil pH (electrochemical probe), nitrogen/phosphorus/potassium concentrations (NPK sensor module), soil temperature (thermistor), and ambient humidity. Data is transmitted via MQTT protocol over Wi-Fi/4G to the cloud backend in real time.

Satellite and Remote Sensing Integration

Sentinel-2 satellite imagery (freely available via the Copernicus Open Access Hub) is processed to compute NDVI and NDWI (Normalized Difference Water Index) values for each registered field. A scheduled pipeline downloads, clips, and processes new imagery at 5-day intervals, feeding computed indices into the prediction models alongside ground sensor data.

Weighted Scoring Algorithm for Soil Classification

A domain-expert-designed weighted scoring algorithm combines multiple sensor readings and computes a composite Soil Health Index (SHI) score on a 0–100 scale. Each parameter (pH, moisture, N, P, K, temperature) is assigned a weight based on its relative impact on rice growth and scored against empirically validated optimal range thresholds. The SHI score maps to three classification levels: Optimal (75–100), Moderate Stress (40–74), and Critical (<40).

Machine Learning Prediction Engine

A Random Forest Regressor (primary model) and an LSTM neural network (time-series model) are trained on a combined dataset of historical field sensor readings, weather data, and recorded yields from Sri Lanka Department of Agriculture datasets. The ensemble prediction outputs nutrient depletion forecasts (7-day horizon), waterlogging probability scores, and estimated days-to-harvest. Models are retrained monthly as new field data accumulates. TensorFlow Lite quantization enables on-device inference for offline use.

Rule-Based Crop Advisory Engine

A structured decision tree of expert-defined if-then rules translates SHI scores and ML predictions into prioritized, plain-language advisory messages. Rules are encoded in a JSON

Classification: Public-SLIIT

ruleset, validated by certified agronomists, and versioned in the system. Examples: if SHI < 40 AND soil moisture < 25% THEN trigger "Critical Irrigation Alert"; if predicted_N_depletion > 30% in 7 days THEN recommend "Apply 50 kg/ha urea within 3 days."

Adaptive Feedback Loop

Post-intervention sensor readings are compared against pre-intervention baselines to assess the effectiveness of applied recommendations. A feedback scoring mechanism identifies advisory rules that consistently fail to produce expected soil responses and flags them for agronomist review, enabling continuous improvement of the advisory rule set over time.

Tools and Platforms with Justification

The mobile application is developed using React Native, which allows the system to run on both Android and iOS devices while maintaining a single codebase. This framework supports reusable user interface components and enables the development of visual, farmer-friendly soil health dashboards and advisory screens suitable for low-literacy users in rural environments.

Sensor data visualisation and interactive field map components are implemented using React Native Maps and Chart.js integrations, which allow the application to display real-time soil health indicators, NDVI overlays, and historical trend charts in a clear and intuitive format.

The backend of the system is developed using Node.js with the Express framework. This platform is chosen because its event-driven architecture efficiently handles multiple concurrent sensor data streams and processes real-time field readings without delays.

The system uses a hybrid database architecture:

- MongoDB is used to store farmer profiles, field sensor readings, prediction history, and advisory logs. Its flexible document-based structure allows easy storage of heterogeneous multi-sensor data collected from different field environments.
- Firebase supports real-time synchronisation of live dashboard updates and progress tracking across sessions. Firebase Authentication is also used to provide secure login access for farmers, agronomists, and field technicians.

This combined architecture provides both scalability and real-time data monitoring, which are essential for an adaptive soil intelligence platform serving thousands of registered fields simultaneously.

The machine learning components are built using Python with scikit-learn for Random Forest and Gradient Boosting models, and TensorFlow Lite for mobile-optimised neural network inference. This allows prediction models to run efficiently on low-cost Android devices even in areas with limited connectivity.

IoT sensor data is transmitted from the rover to the cloud backend using the MQTT protocol, which is a lightweight publish-subscribe protocol specifically designed for low-latency, low-bandwidth environments typical of rural Sri Lankan rice fields.

Classification: Public-SLIIT

Satellite imagery is retrieved and processed through the Sentinel-2 dataset via the Google Earth Engine API, which automates NDVI and NDWI computation at five-day intervals for each registered field.

SMS-based alerts are delivered to farmers without smartphones using the Twilio API, ensuring that time-critical advisory notifications reach all users regardless of their device or internet access.

For development and collaboration, Git is used for version control, with the project repository hosted on GitHub. The system is developed using Visual Studio Code (VS Code), which serves as the main development environment for coding, debugging, and project management.

Data Collection and Ethical Considerations

Data for this research will be collected from three primary sources:

- **Expert Consultation:** Structured interviews with certified agronomists, soil scientists, and agricultural extension officers from the Department of Agriculture Sri Lanka to validate soil classification thresholds and advisory rules.
- **Field Sensor Data:** Real-time and historical sensor readings collected via the rover-mounted IoT sensor array from pilot rice fields in the Kurunegala district.
- **Secondary Datasets:** Historical soil and yield records from the Department of Agriculture Sri Lanka, and publicly available weather data from the Meteorological Department of Sri Lanka, for ML model training.

All participating farmers will provide informed written consent before any data collection. No personally identifiable information will be stored — farmer profiles will be referenced by anonymized field IDs only. All data transfers will be encrypted using HTTPS/TLS. Research activities involving human participants will follow SLIIT ethical research guidelines, and appropriate ethical approval will be obtained prior to field trials.

Validation Strategy and Performance Metrics

Once the system is developed, it will be tested with a small pilot group of approximately 8–10 rice farming households in the Kurunegala district. This pilot study aims to evaluate the system's technical accuracy, usability, and real-world agricultural impact.

Technical Validation

The system's ability to classify soil health and generate appropriate advisory recommendations will be compared against laboratory soil analysis results and expert agronomist assessments to ensure the accuracy of the Soil Health Index scoring algorithm and the ML prediction models.

Intervention Evaluation

The effectiveness of the soil advisory recommendations will be measured by comparing pre-intervention and post-intervention sensor readings across multiple field sessions. Improvements will be assessed in areas such as:

Classification: Public-SLIIT

- Soil pH stability within optimal range for rice cultivation
- Nitrogen, phosphorus, and potassium levels following fertilizer recommendations
- Soil moisture levels following irrigation advisory compliance
- Crop health improvement measured through NDVI trend analysis

Usability Evaluation

Usability will be evaluated through several methods, including observing farmer interactions with the mobile application during field sessions, collecting feedback from participating farmers and agronomists, and measuring usability using the System Usability Scale (SUS).

Performance Targets

The system aims to achieve the following outcomes:

- More than 85% accuracy in classifying soil health levels compared to laboratory reference measurements
- Strong consistency of at least 80% between system-generated advisory recommendations and expert agronomist evaluations
- Measurable improvement in soil nutrient levels and moisture management across pilot fields after the intervention period
- ML prediction models achieving an R^2 score or F1-score of at least 82% on held-out validation data
- High usability among farmers with a System Usability Scale score of 75 or above
- API response time consistently below 2 seconds under normal field usage conditions

4.HIGH-LEVEL SYSTEM ARCHITECTURE

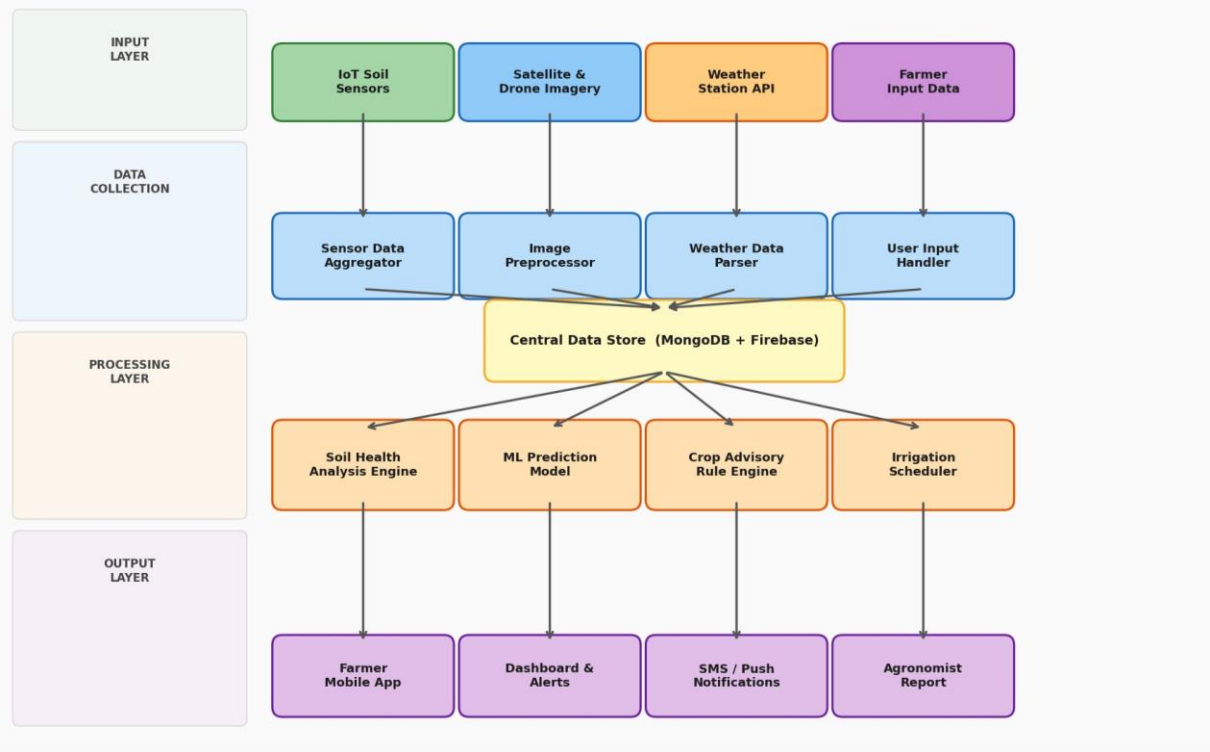


Figure 01 – High Level System Architecture

Classification: Public-SLIIT

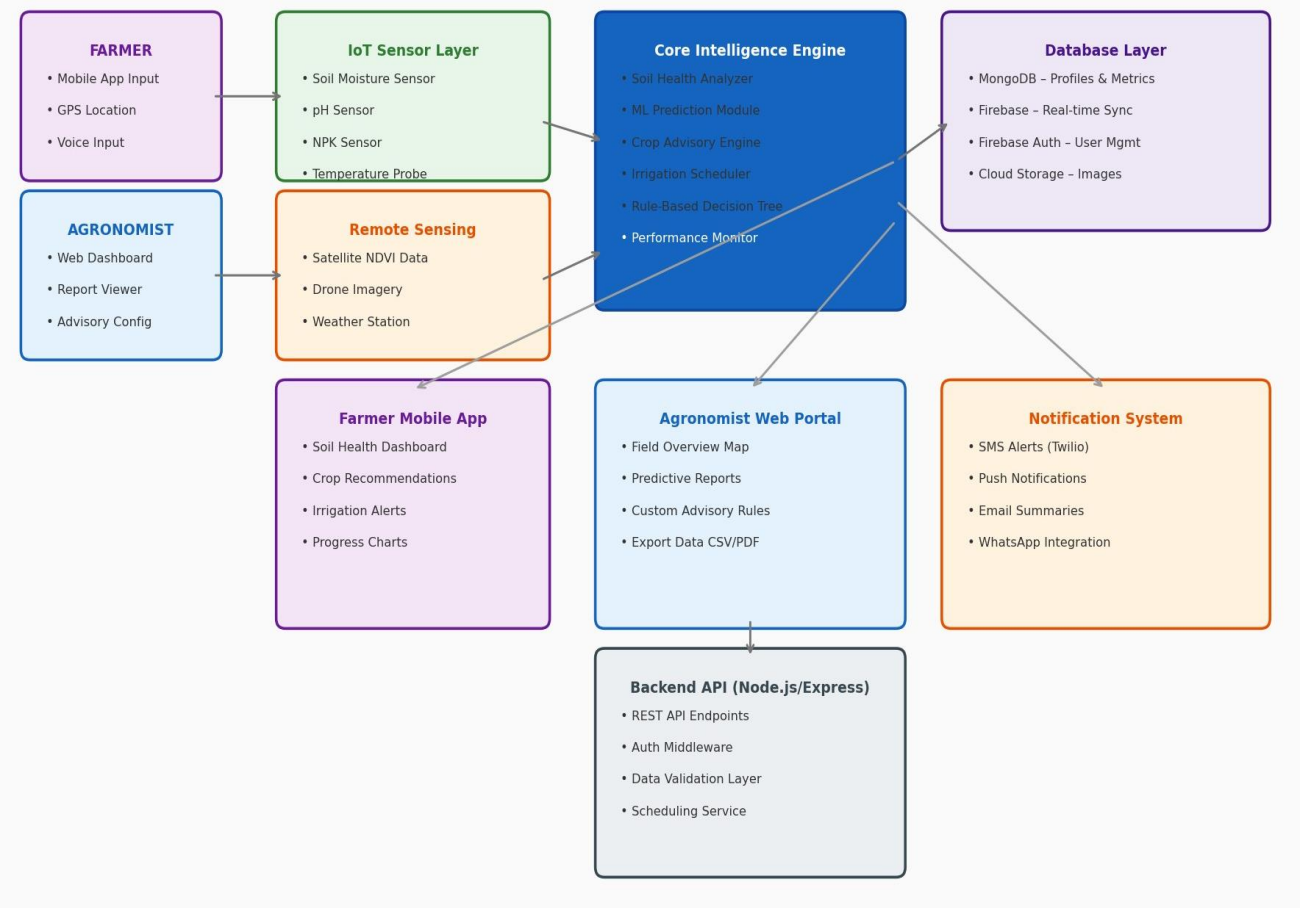


Figure 02 - Component diagram

5.USER REQUIREMENTS

Stakeholder identification

Farmers (Primary Users)

Smallholder rice farmers aged 25–70 who will use the mobile application to receive soil health status, crop advisory notifications, irrigation alerts, and progress tracking. Farmers interact with the system primarily through the mobile app, often on low-cost Android smartphones with 3G/4G connectivity.

Agronomists (Secondary Users)

Certified agricultural specialists who will access the web portal to review field-level soil intelligence, configure advisory rules, validate ML predictions, and generate exportable reports for individual farmers or groups of fields under their advisory mandate.

Rover Operators / Field Technicians (Tertiary Users)

Field personnel responsible for deploying and navigating the autonomous ground rover across rice fields. They interact with the system to initiate scanning sessions and verify sensor connectivity.

Classification: Public-SLIIT

System Administrators

IT personnel responsible for maintaining the backend infrastructure, managing user accounts, and monitoring system health via the admin panel.

Functional Requirements

The system must support several core functions to collect, analyse, and act upon soil health data in real time.

1. Secure User Registration and Login The system should allow farmers, agronomists, and rover operators to create user accounts and securely log into the mobile application and web portal using Firebase Authentication.

2. Rover Sensor Data Ingestion The system should receive real-time soil sensor data from rover-mounted IoT sensors via MQTT protocol and store structured readings (pH, moisture, NPK, temperature) in MongoDB.

3. Satellite Imagery Processing The system should automatically retrieve and process Sentinel-2 satellite imagery for registered fields at 5-day intervals and compute NDVI and NDWI indices for crop health monitoring.

4. Soil Health Index Calculation The system should compute a composite Soil Health Index (SHI) score from multi-sensor data and classify field soil health into three levels: Optimal, Moderate Stress, or Critical.

5. ML-Based Predictive Analysis The system should run trained machine learning models to generate 7–14 day forecasts of soil nutrient depletion, waterlogging risk, and optimal harvest windows based on sensor and weather data.

6. Crop Advisory Generation The system should translate SHI scores and ML prediction outputs into prioritized, plain-language crop advisory recommendations tailored to the farmer's specific field conditions.

7. Alert Notification System The system should send push notifications (via Firebase Cloud Messaging) and SMS alerts (via Twilio) for time-critical field events such as waterlogging risk or urgent fertilizer needs.

8. Farmer Mobile Dashboard The system should display soil health status, advisory messages, field maps, and historical trend charts through a visual, low-text mobile interface accessible in Sinhala and English.

9. Agronomist Web Dashboard The system should provide agronomists with field-level soil intelligence maps, longitudinal soil trend charts, prediction reports, and PDF/CSV data export functionality.

10. Adaptive Feedback Logging The system should log post-intervention sensor readings and score the effectiveness of applied recommendations to support continuous improvement of the advisory rule engine.

Classification: Public-SLIIT

Non-Functional Requirements

Non-functional requirements describe how well the system performs and the quality attributes it must maintain.

Usability The mobile application must present soil health data and advisory recommendations through a simple, visual, and intuitive interface so that farmers with limited technical literacy can interact with it comfortably in both Sinhala and English.

Performance The system should respond within 2 seconds during dashboard loading and advisory generation activities to ensure smooth interaction without delays. Real-time sensor data must be ingested and processed with less than 5 seconds of latency.

Security User data, including farmer profiles, field sensor readings, and soil health records, must be securely stored and protected from unauthorized access. All data transfers must be encrypted using HTTPS/TLS, and field data must be anonymised using unique field IDs.

Reliability The system should operate consistently without crashes or data loss during active field monitoring sessions. The mobile application must support offline caching so that farmers can access their last-fetched recommendations even without internet connectivity.

Accessibility The application should be accessible on common low-cost Android smartphones (Android 8.0 and above) to ensure easy use in both rural farm environments and agricultural extension offices. SMS-based alerts must serve as a fallback for users in areas with no data connectivity.

Scalability The system architecture should allow future expansion, such as supporting additional soil sensor types, integrating new crop varieties, adding more advisory rules, or scaling to support thousands of registered fields across multiple districts.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system is smallholder rice farmers in Sri Lanka who face challenges in managing soil health, fertilizer application, and irrigation decisions without access to affordable, real-time field intelligence tools. Early adoption of precision agriculture technology during the growing season is important because soil nutrient levels and moisture conditions change rapidly and require timely intervention.

The main customers who would adopt the system include:

Smallholder Rice Farmers: Farmers operating 0.5–2 hectares of paddy land who want data-driven guidance on fertilizer scheduling, irrigation timing, and crop health monitoring will use the mobile application to receive soil health assessments and actionable advisory recommendations directly on their smartphones.

Agricultural Extension Officers and Agronomists: Government extension officers and certified agronomists who advise farming communities across Sri Lanka's major rice-growing districts may

Classification: Public-SLIIT

use the web dashboard as a supportive decision tool to monitor multiple fields simultaneously and provide more accurate, data-backed recommendations to the farmers under their care.

Agricultural Input Retailers and Cooperatives: Fertilizer retailers and farmer cooperatives may use the platform to offer value-added advisory services to their customers, improving fertilizer sales accuracy and building stronger relationships with farming households.

Government Agricultural Agencies: The Department of Agriculture Sri Lanka and provincial agricultural ministries may adopt the system as a scalable digital extension tool to improve national paddy yield statistics and reduce post-harvest losses across the country.

The global precision agriculture market is projected to reach USD 12.8 billion by 2027, with the highest growth rates observed in Asia-Pacific developing economies [14]. In Sri Lanka alone, rice cultivation supports approximately 1.8 million farming households, representing a substantial addressable market for affordable, mobile-first soil intelligence solutions.

Value Proposition

Value Element	Description
Affordable Soil Intelligence	Replaces expensive laboratory soil testing (LKR 3,000–8,000 per test) with continuous real-time monitoring at a fraction of the cost.
Actionable Advisory — Not Just Data	Translates complex sensor and satellite data into simple, ranked, farmer-language recommendations — no agronomic expertise required.
Timely Alerts	SMS and push notifications for waterlogging, nutrient depletion, and irrigation windows reduce crop loss from delayed action.
Reduced Fertilizer Waste	Precision NPK recommendations reduce over-fertilization by an estimated 15–25%, directly reducing input costs for farmers.
Progress Visibility	Longitudinal soil health trend charts enable farmers and agronomists to track field improvement over multiple seasons.
Inclusive Design	Sinhala-language interface, visual-first design, and SMS fallback ensure accessibility for low-literacy and low-connectivity users.
Offline Capability	Cached recommendations and local data storage enable continued use in areas with intermittent internet connectivity.

Table 02 - Value Proposition

Classification: Public-SLIIT

Revenue Model

1. Freemium Subscription (B2C)

Free tier: basic soil health dashboard and weekly advisory summaries. Premium tier (LKR 500–800/month): real-time alerts, full prediction reports, unlimited advisory history, and priority SMS notifications.

2. Institutional Licensing (B2G)

Annual licensing to the Department of Agriculture for deployment to agricultural extension networks across Sri Lanka's rice-growing districts.

3. B2B Data Services

Anonymized, aggregated soil intelligence data sold to Agri-input companies, agricultural insurers, and commodity traders for market forecasting.

4. White-Label Platform Licensing

License the platform to NGOs and international agricultural development organizations for adaptation to other tropical crops and countries.

Cost Estimation

Cost Category	Estimated Cost (LKR)	Details
Software Development	Rs. 40,000 – 60,000	React Native app, Node.js backend, ML model integration, web dashboard
IoT Sensor Hardware (Pilot)	Rs. 25,000 – 40,000	Soil moisture, pH, NPK, and temperature sensors for rover prototype
Cloud Infrastructure	Rs. 8,000 – 12,000	MongoDB Atlas, Firebase, cloud compute for ML model hosting (6 months)
Expert Consultation	Rs. 10,000 – 15,000	Agronomist fees for advisory rule validation and system review
Pilot Field Testing	Rs. 12,000 – 18,000	Farmer recruitment, field visits, lab reference soil tests, usability testing
TOTAL INITIAL COSTS	Rs. 95,000 – 145,000	

Table 03 - Cost Estimation

Classification: Public-SLIIT

Competitive Advantage

The proposed system offers several distinct advantages over existing precision agriculture and soil monitoring tools available in Sri Lanka and globally:

- Full integration of IoT sensor data, satellite imagery, weather feeds, and farmer observations in a single unified platform — no existing system in Sri Lanka offers this.
- Closed-loop adaptive advisory system that learns from field outcomes and continuously improves recommendation quality.
- Rover-mounted sensor support enabling spatially resolved soil mapping without fixed sensor grid installation costs.
- Localised for Sri Lankan tropical rice varieties, soil types, and the Department of Agriculture's established agronomy guidelines.

Intellectual Property Considerations

The intellectual property of the proposed system encompasses the software codebase, trained ML model weights, the Soil Health Index scoring algorithm, and the advisory rule knowledge base. The following IP protection measures are considered:

- Copyright registration for all software source code and model architectures.
- Documentation and filing of the Soil Health Index (SHI) algorithm and weighted scoring methodology as a novel computational contribution.
- Trademark registration of the platform name and branding upon commercialization.
- Data licensing agreements protecting the privacy of farmer field data while enabling aggregate data monetization.

7.BUDGET AND JUSTIFICATION

Cost Category	Estimated Cost (LKR)	Justification
Travel Expenses	Rs. 12,000	Visits to SLIIT Malabe campus for supervisor meetings; field visits to pilot rice fields in Kurunegala district; visits to Department of Agriculture offices for data and expert access.
Data Collection	Rs. 15,000	Private appointments with agronomists and soil scientists; laboratory reference soil testing (3–5 samples) for ML model validation; farmer usability trial incentives.
Sensor Hardware (Prototype)	Rs. 18,000	Purchase of NPK sensor module, pH probe, capacitive moisture sensor, and temperature sensor for integration with rover prototype.
Printing & Documentation	Rs. 2,500	Printing of proposal reports, consent forms, expert questionnaires, and field data collection sheets.
TOTAL	**Rs. 47,500**	

Table 04 - Budget and Justification

Classification: Public-SLIIT

8.WORK BREAKDOWN STRUCTURE (WBS)

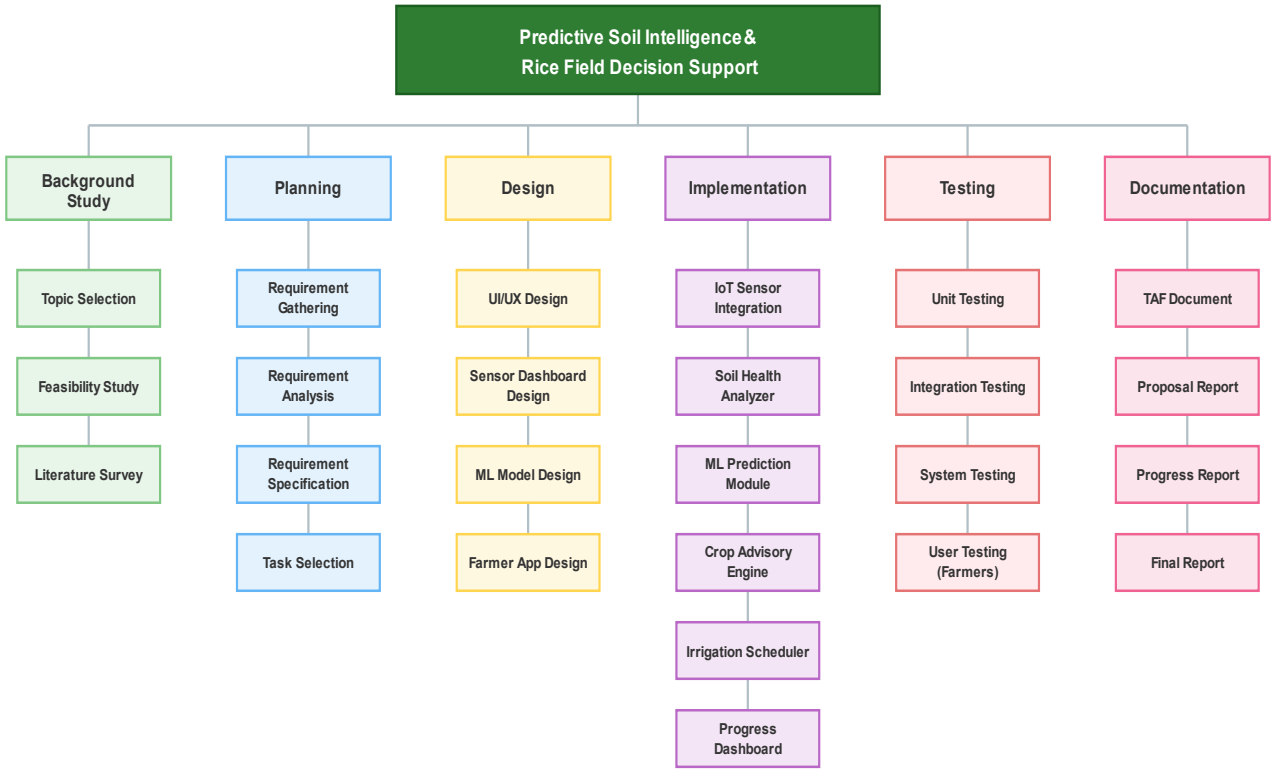


Figure 03 - Work Breakdown Structure

9. GANTT CHART

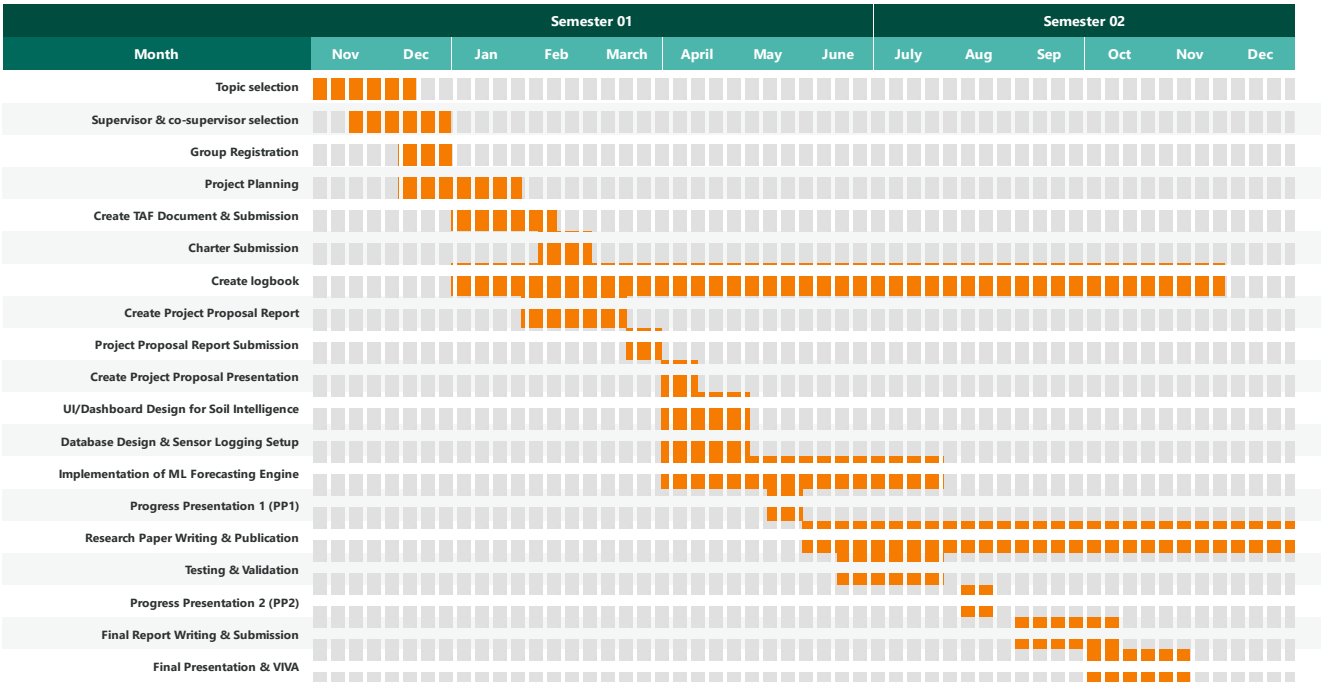


Table 04 - Gantt Chart

Classification: Public-SLIIT

10 .REFERENCES LIST

[1] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of Things (IoT): A vision, architectural elements, and future directions," *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645–1660, 2013.

[2] M. D. Pathiraja, D. Griffith, D. Farrell, and P. Danaher, "Smallholder agriculture in developing countries: A systematic review of barriers and opportunities," *World Development*, vol. 115, pp. 42–57, 2019.

[3] Q. Zhang, *Precision agriculture technology for crop farming*. CRC Press, 2016.

[4] S. Wolfert, L. Ge, C. Verdouw, and M. J. Bogaardt, "Big data in smart farming: A review," *Agricultural Systems*, vol. 153, pp. 69–80, 2017.

[5] United Nations, "Transforming our world: The 2030 Agenda for Sustainable Development," UN General Assembly, 2015.

[6] A. Kamilaris and F. X. Prenafeta-Boldu, "Deep learning in agriculture: A survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.

[7] X. E. Pantazi, D. Moshou, T. Alexandridis, R. L. Whetton, and A. M. Mouazen, "Wheat yield prediction using machine learning and advanced sensing techniques," *Computers and Electronics in Agriculture*, vol. 121, pp. 57–65, 2016.

[8] N. Nevavuori, N. Narra, and T. Lipping, "Crop yield prediction with deep convolutional neural networks," *Computers and Electronics in Agriculture*, vol. 163, pp. 1–11, 2019.

[9] J. A. Gamon, C. B. Field, M. L. Goulden, K. L. Griffin, A. E. Hartley, et al., "Relationships between NDVI, canopy structure, and photosynthesis in three Californian vegetation types," *Ecological Applications*, vol. 5, no. 1, pp. 28–41, 1995.

[11] A. Bechar and C. Vigneault, "Agricultural robots for field operations: Concepts and components," *Biosystems Engineering*, vol. 149, pp. 94–111, 2016.

[12] J. Schimmelpfennig, "Farm profits and adoption of precision agriculture," U.S. Department of Agriculture, Economic Research Service, EIB-217, 2016.

[13] MarketsandMarkets, "Precision agriculture market by technology, application, and geography — global forecast to 2027," 2022. [Online]. Available: https://www.marketsandmarkets.com

[14] Department of Agriculture Sri Lanka, "Paddy statistics 2022/2023," Ministry of Agriculture, Sri Lanka, 2023.

Classification: Public-SLIIT

11.APPENDICES

Survey instruments

Predictive Soil Intelligence and Rice Field Decision Support System

Introduction

The Predictive Soil Intelligence and Rice Field Decision Support System is a research and development project conducted by a final-year undergraduate student in Information Technology at the Sri Lanka Institute of Information Technology. The project focuses on designing an intelligent, sensor-driven mobile platform that monitors real-time soil health conditions in rice fields, predicts soil nutrient depletion and waterlogging risk, and delivers actionable crop advisory recommendations to smallholder rice farmers in Sri Lanka.

The purpose of this questionnaire is to gather professional insights from agronomists, soil scientists, and agricultural extension officers about current soil assessment practices, fertilizer recommendation methods, and effective field management strategies for rice cultivation in Sri Lanka's major paddy-growing regions.

The information collected will be used only for academic research and system development purposes. It will help design a more accurate, practical, and farmer-friendly soil intelligence platform that reflects real-world agricultural conditions and expert knowledge.

All responses will be kept confidential and will only be used to improve the quality, usability, and effectiveness of the proposed system.

Questions

1.What are the most critical soil parameters to monitor for optimal rice cultivation in Sri Lanka’s major paddy-growing regions (Kurunegala, Polonnaruwa, Anuradhapura)?

2.What are the commonly recommended optimal ranges for soil pH, nitrogen, phosphorus, potassium, and moisture for Sri Lankan rice varieties (e.g., Bg 403, Bg 250)?

3. How frequently should soil parameters ideally be measured during a single rice cultivation season?

4.What are the most common soil-related problems that reduce rice yields in Sri Lanka, and which are most often mismanaged by smallholder farmers?

5.What types of fertilizer recommendations do you currently provide to farmers, and how do you determine the timing and quantity?

6.Are there established methods to classify field soil health into severity levels (e.g., optimal, moderate stress, critical)? If so, what thresholds do you use?

7.What indicators (observable or measurable) signal that a farmer should take immediate action to prevent crop loss?

Classification: Public-SLIIT

8.What information do farmers most commonly request when they contact extension services about soil or crop problems?

9.If a mobile application were to provide automated soil health assessments and crop advisory messages, what features would make it most useful and trustworthy for farmers?

10.What are the signs that a soil advisory recommendation has been followed correctly, and how would you measure its effectiveness?

11.Are there seasonal, regional, or variety-specific factors that should adjust soil management recommendations significantly?

12.What types of visual progress reports or dashboards would be most useful for agronomists monitoring multiple farms simultaneously?

13.From your professional experience, what is one feature you wish precision agriculture tools designed for Sri Lankan rice farmers did better?

14.Would you be willing to participate in a validation review of the system's generated advisory recommendations once the prototype is developed?

Risk analysis

Inaccurate Soil Sensor Readings (sensor drift or calibration error)	High	Medium	Regular sensor calibration against lab reference; outlier detection in data pipeline; alerts for abnormal reading spikes.
Poor ML Model Accuracy on Local Data	High	Medium	Train on locally collected Sri Lankan paddy field data; cross-validate with agronomist-verified ground truth; retrain monthly as data accumulates.
Farmer Non-Adoption of Mobile App	Medium	Medium	Co-design UI with farmer focus groups; offer SMS fallback; provide training sessions through agricultural extension officers.
Internet Connectivity Gaps in Rural Areas	Medium	High	Implement local data caching on mobile app; SMS notification fallback via Twilio; offline mode with last-synced recommendations.
Data Privacy and Security Breach	Critical	Low	HTTPS/TLS encryption; Firebase Auth; field data anonymised by ID; no personal data stored on mobile device.
Satellite Imagery Unavailability (cloud cover)	Medium	High	Use multi-spectral composite imagery; fall back to ground sensor data only when NDVI unavailable; flag data gaps to agronomist.
Advisory Recommendation Errors	High	Low	Expert agronomist review of all advisory rules before deployment; confidence scores displayed alongside recommendations; agronomist override capability.

Table 06-Risk Analysis

Classification: Public-SLIIT